

Cyclotomic realization by topological Hochschild homology

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Abstract

We study the essential image of topological Hochschild homology on small idempotent-complete stable ∞ -categories. We construct a genuine cyclotomic THH functor by Morita descent and compare it naturally with Nikolaus–Scholze THH. Every cyclotomic spectrum whose underlying spectrum is bounded below is realizable by THH. The proof combines corepresentability of integral TR with a retract construction for arbitrary coproducts of reduced polynomial motives and a convergent cellular tower. In the unbounded case there are two distinct failures of realization. An Anderson-dual example has no genuine cyclotomic refinement. A second example admits such a refinement but violates the uniform prime-wise estimate supplied by the integer May filtration on THH. Consequently the essential image of THH is strictly smaller than the essential image of genuine cyclotomic spectra in the Nikolaus–Scholze category, and the latter is itself a proper subcategory.

Contents

1	Introduction	2
2	Cyclotomic categories and conventions	3
3	The genuine cyclotomic THH functor	5
4	Realization of bounded-below cyclotomic spectra	13
5	A cyclotomic spectrum with no genuine refinement	20
6	An obstruction across the primes	22
7	Essential images and the role of boundedness	28
A	Anderson duality and iterated Tate constructions	30
	References	34

1 Introduction

Topological Hochschild homology assigns to a small idempotent-complete stable ∞ -category a spectrum with a circle action and cyclotomic Frobenius maps. We study the converse problem: which cyclotomic spectra occur in this way? The answer depends on boundedness and on the distinction between genuine cyclotomic spectra and cyclotomic spectra in the sense of Nikolaus–Scholze.

Write Cat^{perf} for the ∞ -category of small idempotent-complete stable ∞ -categories and exact functors. Let $\text{CycSp}^{\text{gen}}$ and CycSp_{NS} denote the genuine and Nikolaus–Scholze cyclotomic ∞ -categories, respectively, and let

$$U : \text{CycSp}^{\text{gen}} \longrightarrow \text{CycSp}_{\text{NS}}$$

be the comparison functor of [13, Proposition II.3.4]. An object of either category is called *bounded below* if its underlying spectrum is bounded below. The comparison restricts to an equivalence on these full subcategories [13, Theorem II.6.9].

Theorem 1.1. *The following statements hold.*

(i) *There is an ∞ -functor*

$$\text{THH}^{\text{gen}} : \text{Cat}^{\text{perf}} \longrightarrow \text{CycSp}^{\text{gen}}$$

and a natural equivalence $U\text{THH}^{\text{gen}} \simeq \text{THH}_{\text{NS}}$ in $\text{Fun}(\text{Cat}^{\text{perf}}, \text{CycSp}_{\text{NS}})$, where THH_{NS} denotes the usual cyclotomic topological Hochschild homology.

(ii) *Every bounded-below object of CycSp_{NS} is equivalent to $\text{THH}_{\text{NS}}(\mathcal{C})$ for some $\mathcal{C} \in \text{Cat}^{\text{perf}}$. Every bounded-below object of $\text{CycSp}^{\text{gen}}$ is likewise equivalent to $\text{THH}^{\text{gen}}(\mathcal{C})$ for some $\mathcal{C} \in \text{Cat}^{\text{perf}}$.*

(iii) *As full subcategories of CycSp_{NS} , the essential images satisfy*

$$\text{EssIm}(\text{THH}_{\text{NS}}) \subsetneq \text{EssIm}(U) \subsetneq \text{CycSp}_{\text{NS}}. \quad (1)$$

In particular, THH^{gen} is not essentially surjective.

The realization in part (ii) allows arbitrary small stable categories. The connective ring spectra used in its proof provide mapping comparisons in localizing motives; the final category is obtained from a realization theorem for motives. No connectivity condition on its mapping spectra is part of the conclusion.

The two strict inclusions in (1) have different witnesses. Fix a prime p , and write $T_p Y = \rho_p^*(Y^{tC_p})$ for the prime Tate construction with its residual circle action, reindexed by the root isomorphism $\rho_p : \mathbb{T} \xrightarrow{\sim} \mathbb{T}/C_p$. Let $I_{\mathbb{Z}}$ be the Anderson dual of the sphere, with trivial circle action. The spectrum

$$I_{\mathbb{Z}} \oplus T_p I_{\mathbb{Z}} \oplus T_p^2 I_{\mathbb{Z}}$$

with shift Frobenius at p and zero Frobenius at the other primes has no genuine refinement. Its obstruction is the factorization imposed by the p^2 cyclotomic coherence: the relevant proper Tate construction vanishes, whereas the iterated prime Tate construction does not.

For the first strict inclusion, we construct a cyclotomic structure on the spectrum

$$H\mathbb{Z} \oplus \bigoplus_{p \text{ prime}} \Sigma^{-2p^2} H\mathbb{Z} \quad (2)$$

with trivial circle action. This structure admits a genuine refinement. It nevertheless violates a condition satisfied by every categorical THH: for each fixed homotopy class, all prime Frobenius images factor through Tate constructions of connective covers whose lower bounds decrease at most linearly in p . The cyclotomic maps on (2) detect the summand in degree $-2p^2$ for every

prime p . Thus the obstruction applies to its NS cyclotomic structure itself, and consequently to any genuine realization by THH.

There are three main steps in the proofs. First, the classical Blumberg–Mandell functor on spectral categories [6, Theorem 4.9] is placed in a precise model of genuine cyclotomic spectra. Morita invariance then allows descent along the localization presenting Cat^{perf} [5, Proposition 4.4]. We construct the comparison with the Tate-diagonal Frobenius as a natural equivalence before descending it. This establishes part (i) at the level of functor ∞ -categories.

Second, Efimov’s comparison between motivic and cyclotomic mapping spectra [8, Section 9.1, equation (9.4), and Proposition 9.4] is extended from connective ring targets to the finite stable closure of arbitrary coproducts of the reduced polynomial motive. The extension uses a connective ring that packages each such coproduct as a retract. McCandless’s corepresentability of integral TR [12, Theorem 3.3.12] supplies maps onto degree-zero homotopy groups. Iterating these maps gives a convergent cellular tower, whose entire diagram lifts to motives. The theorem of Ramzi–Sosnilo–Winges [15, Theorem 1.5] then realizes its colimit by a small stable category.

Third, the filtered Tate diagonal of Antieau–Mathew–Morrow–Nikolaus [2, Appendix A] gives an integer-indexed May filtration on the THH of any associative ring spectrum. Its Frobenius estimate passes to arbitrary small stable categories through finite sets of generators. For (2), the proper-Tate description of genuine cyclotomic spectra [1, Theorem B] supplies the genuine refinement: the higher compatibility targets vanish for the relevant integral module spectra. The May estimate then excludes THH realization.

Section 2 fixes the categorical and Tate conventions. Theorems 3.1 and 4.1 prove parts (i) and (ii). Theorems 5.2 and 6.1 provide the two witnesses in part (iii). The detailed Anderson-duality calculation is given in the appendix and is cited through its numbered statements.

2 Cyclotomic categories and conventions

We fix compatible Grothendieck universes. Stable categories, spectra, and all indexing sets used in a construction are small in the chosen universe; categories of such objects and their functor categories are formed in a larger universe. All limits, colimits, tensor products, and fixed-point constructions are derived unless an explicit point-set model is specified. We write $\text{map}_{\mathcal{D}}(X, Y)$ for the mapping spectrum in a stable ∞ -category, $\text{Map}_{\mathcal{D}}(X, Y)$ for its underlying mapping space, and $[X, Y]_{\mathcal{D}} = \pi_0 \text{map}_{\mathcal{D}}(X, Y)$.

Let $\mathbb{T}$ be the circle group. The category of spectra with a homotopy circle action is $\text{Sp}^{B\mathbb{T}} = \text{Fun}(B\mathbb{T}, \text{Sp})$, also written $\text{Sp}^{h\mathbb{T}}$. For each positive integer m , let $C_m \subset \mathbb{T}$ be the subgroup of order m . We fix the root isomorphism $\rho_m : \mathbb{T} \xrightarrow{\sim} \mathbb{T}/C_m$, inverse to the isomorphism $\mathbb{T}/C_m \rightarrow \mathbb{T}$ induced by the m th power map. Iterated root reindexings are identified by their canonical composition equivalences.

For a finite group G acting on a spectrum Y , the ordinary Tate construction is

$$Y^{tG} = \text{cofib}(Y_{hG} \xrightarrow{\text{Nm}} Y^{hG}).$$

For a normal subgroup, both the norm sequence and its Tate term carry the residual quotient action. In particular, iterating a prime Tate construction retains that residual action. We set

$$T_p Y = \rho_p^*(Y^{tC_p}) \quad (p \text{ prime}).$$

The *proper Tate construction* $Y^{\tau G}$ is the geometric G -fixed-point spectrum of the Borel-complete genuine G -spectrum associated to Y . It uses the family of all proper subgroups of G . For circle spectra it retains a residual $\mathbb{T}/C_m$ -action, and we write

$$S_m Y = \rho_m^*(Y^{\tau C_m}) \quad (m > 1), \quad S_1 = \text{id}.$$

For prime p there is a natural identification $S_p = T_p$; for composite m , proper Tate and ordinary Tate are different constructions.

An object of CycSp_{NS} is a homotopy circle spectrum X together with circle-equivariant maps $\varphi_p : X \rightarrow T_p X$ for all primes p . Morphisms and their higher homotopies are those of the corresponding lax equalizer [13, Definition II.1.6]. We denote the forgetful functor to spectra by $V : \text{CycSp}_{\text{NS}} \rightarrow \text{Sp}$, and use the same letter X for a cyclotomic object and its homotopy circle spectrum when its type is specified by the expression in which it occurs.

The category $\text{CycSp}^{\text{gen}}$ uses genuine circle spectra with equivalences detected at finite subgroups. Its objects carry coherent equivalences $\rho_m^* \Phi^{C_m} E \simeq E$ for all $m \geq 1$; here Φ^{C_m} denotes geometric fixed points. We use the model of [13, Sections II.3 and II.6]. Equivalently, the proper-Tate description of [1, Theorem B and Theorem 2.81] expresses this category as a right-lax limit for the left-lax action by the functors S_m . Besides maps $X \rightarrow S_m X$, this description includes the composition homotopies and all their higher coherences. In the proof of Theorem 6.1 these data will be constructed explicitly.

The comparison $U : \text{CycSp}^{\text{gen}} \rightarrow \text{CycSp}_{\text{NS}}$ sends the genuine structure equivalence to the prime Frobenius by the composite

$$E \xrightarrow{\sim} \rho_p^* \Phi^{C_p} E \longrightarrow \rho_p^*(E^{tC_p})$$

on homotopy circle spectra, where the second arrow is induced by Borel completion [13, Propositions II.3.2 and II.3.4].

Lemma 2.1. *The category CycSp_{NS} is stable and presentable, and V is exact, conservative, and preserves small colimits. The underlying-spectrum functor $V_{\text{gen}} : \text{CycSp}^{\text{gen}} \rightarrow \text{Sp}$ is also exact, conservative, and preserves small colimits.*

Proof. The NS statement is [13, Corollary II.1.7]. The genuine category is stable, being the limit of the geometric-fixed-point action on a stable category by exact functors. For its colimits, form the colimit of a diagram in genuine circle spectra with finite-isotropy equivalences. Geometric fixed points preserve colimits. The given compatible equivalences $\rho_m^* \Phi^{C_m} E_i \simeq E_i$ therefore induce such equivalences on the colimit, together with their coherences. This construction has the universal property of the colimit in the category of coherent fixed objects: its mapping-space description follows by taking limits of the compatible mapping-space descriptions at each stage. This is the colimit construction for the geometric-fixed-point action in [13, Sections II.5–II.6]. The genuine-to-underlying-spectrum functor preserves these colimits and finite limits, proving exactness. Finally, if a genuine cyclotomic object's underlying spectrum vanishes, all its finite geometric fixed points vanish by its cyclotomic equivalences. These fixed-point functors jointly detect the zero object. Exactness now gives conservativity on all morphisms. $\square$

The pointwise t -structure on $\text{Sp}^{B\mathbb{T}}$ is used for all connective covers below. Thus $\tau_{\geq a} Y$ means the connective cover as a homotopy circle spectrum, and its underlying spectrum is the usual connective cover. If $Y \in \text{Sp}_{\geq a}^{B\mathbb{T}}$ and $Z \in \text{Sp}_{\leq a-1}^{B\mathbb{T}}$, then

$$\text{Map}_{\text{Sp}^{B\mathbb{T}}}(Y, Z) \simeq *. \quad (3)$$

Indeed, the underlying mapping space is contractible by t -structure orthogonality, and taking its homotopy fixed points remains contractible. This is an equivariant nullhomotopy statement, and it can consequently be used before applying a Tate functor.

For $X \in \text{CycSp}_{\text{NS}}$, the notation $X \in (\text{CycSp}_{\text{NS}})_{\geq a}$ will mean only $VX \in \text{Sp}_{\geq a}$. No cyclotomic connective-cover functor is implicit in this notation. The analogous convention applies to genuine objects. In these terms the bounded-below comparison is the equivalence

$$U : (\text{CycSp}^{\text{gen}})_{\text{bb}} \xrightarrow{\sim} (\text{CycSp}_{\text{NS}})_{\text{bb}} \quad (4)$$

of [13, Theorem II.6.9].

Unless a different refinement is indicated, THH denotes THH_{NS} , or its underlying spectrum or circle spectrum when the target category is specified.

3 The genuine cyclotomic THH functor

Theorem 3.1. *There is an ∞ -functor*

$$\mathrm{THH}^{\mathrm{gen}} : \mathrm{Cat}^{\mathrm{perf}} \longrightarrow \mathrm{CycSp}^{\mathrm{gen}}$$

and a natural equivalence

$$U\mathrm{THH}^{\mathrm{gen}} \simeq \mathrm{THH}_{\mathrm{NS}} \quad \text{in } \mathrm{Fun}(\mathrm{Cat}^{\mathrm{perf}}, \mathrm{CycSp}_{\mathrm{NS}}).$$

The construction has three stages. We first turn the spacewise cyclotomic structure of Blumberg–Mandell into the precise point-set structure used by Nikolaus–Scholze. This gives a functor on strict spectral categories. We then prove that this functor inverts Morita equivalences and descend it by the universal property of localization. Finally, a comparison on the total category of labelled cyclic words identifies its image under U with the cyclic-bar construction and its Tate-diagonal Frobenius maps. The last comparison takes place in a functor ∞ -category and includes the natural homotopies required by the cyclotomic structures.

3.1 The source presentation

Let $\mathbf{S}$ be the ordinary category of small categories enriched in symmetric spectra of pointed simplicial sets whose mapping spectra are cofibrant, with all strict enriched functors as morphisms. An enriched functor is a *Morita equivalence* if derived extension of scalars induces an equivalence of categories of perfect modules. Write W_{Mor} for these morphisms. The spectral Morita presentation [5, Proposition 4.4] gives

$$q : \mathbf{NS} \longrightarrow \mathrm{Cat}^{\mathrm{perf}}, \quad q(\mathcal{A}) = \mathrm{Perf}(\mathcal{A}), \quad (\mathbf{NS})[W_{\mathrm{Mor}}^{-1}] \simeq \mathrm{Cat}^{\mathrm{perf}}. \quad (5)$$

The pointwise cofibrant replacement used in this presentation is supplied by [5, Proposition 4.1].

We use the cyclic-bar definition of $\mathrm{THH}_{\mathrm{NS}}$: on a spectral category one forms the derived cyclic bar and equips it with the prime Frobenius maps induced by the multivariable Tate diagonal. The construction below proves that this cyclotomic functor inverts Morita equivalences and therefore descends along q . In particular, Morita invariance of this definition is not an input to the comparison.

3.2 From spacewise structure maps to genuine cyclotomic spectra

Use the complete circle universe of [13, Section II.3],

$$\mathcal{U} = \bigoplus_{k \in \mathbb{Z}, i \geq 1} \mathbb{C}_{k,i},$$

where $z \in \mathbb{T}$ acts on $\mathbb{C}_{k,i}$ by z^k . Let $\rho_n : \mathbb{T} \cong \mathbb{T}/C_n$ be the inverse of the n th-power isomorphism. Equip the fixed universes with the isometries

$$\alpha_n : \rho_n^* \mathcal{U}^{C_n} \xrightarrow{\cong} \mathcal{U}, \quad \mathbb{C}_{n\ell,i} \longmapsto \mathbb{C}_{\ell,i},$$

which are the identity on each indicated complex line. Under the canonical identification of iterated root pullbacks and fixed subspaces, they satisfy the equality

$$\alpha_n \circ \rho_n^* (\alpha_m^{C_n}) = \alpha_{mn} : \rho_{mn}^* \mathcal{U}^{C_{mn}} \longrightarrow \mathcal{U}.$$

Both sides send the line indexed by $(mn\ell, i)$ identically to the line indexed by (ℓ, i) . We use these relabelings throughout the homotopy-colimit geometric-fixed-point model of [13, Definition II.2.10], and put

$$D_n = \rho_n^* \Phi_{\mathcal{U}}^{C_n}.$$

On a trivial representation its point-set formula is

$$(D_n T)(\mathbb{R}^k) = \rho_n^* \operatorname{hocolim}_{\substack{W \subset \mathcal{U} \\ W^{C_n} = 0}} T(\mathbb{R}^k \oplus W)^{C_n}. \quad (6)$$

We use the specified Bousfield–Kan bar construction for this homotopy colimit. The model has natural comparison maps

$$\mu_{n,m} : D_n D_m \longrightarrow D_{mn},$$

which are equivalences by [13, Proposition II.2.12]. Here α_m^{-1} identifies the outer indexing universe for $D_n D_m$ with $\rho_m^* \mathcal{U}^{C_m}$; the displayed compatibility makes its composite relabeling agree with that for D_{mn} .

Let $\mathbf{Cyc}_{\mathbf{BM}}$ denote the ordinary category in [6, Definition 4.2]. An object is an orthogonal circle spectrum T with maps

$$r_{n,L} : \rho_n^* T(L)^{C_n} \longrightarrow T(\rho_n^* L^{C_n})$$

for finite-dimensional circle representations L . The axioms require the unit identity, compatibility with the orthogonal-spectrum structure maps, the strict iterated-restriction identity in condition (iii), and the isomorphisms on geometric stable homotopy groups in condition (iv). Let $\mathbf{Cyc}_{\mathbf{NS}}^O$ denote the point-set category of [13, Definition II.3.6].

Lemma 3.2. *There is an ordinary functor*

$$\Xi : \mathbf{Cyc}_{\mathbf{BM}} \longrightarrow \mathbf{Cyc}_{\mathbf{NS}}^O$$

which preserves the underlying orthogonal circle spectra and their maps.

Proof. For each index W of (6), the map $r_{n,\mathbb{R}^k \oplus W}$ has target $T(\mathbb{R}^k)$. Condition (ii) of [6, Definition 4.2], applied to an inclusion of indexing representations and to the zero vector in its orthogonal complement, says that these maps form a strictly compatible cocone. On every simplex of the Bousfield–Kan construction, apply the cocone map and collapse the simplex coordinate. This defines a map

$$R_n : D_n T \longrightarrow T.$$

The same condition makes it compatible with the $O(k)$ -actions and the orthogonal-spectrum structure maps.

We claim the equality of point-set maps

$$R_n \circ D_n(R_m) = R_{mn} \circ \mu_{n,m}. \quad (7)$$

In the construction of $\mu_{n,m}$ in [13, Proposition II.2.12], the iterated homotopy colimit is expressed by the double bar with terms

$$T(\mathbb{R}^k \oplus W \oplus V)^{C_{mn}}, \quad W \subset \mathcal{U}^{C_m}, \quad W^{C_{mn}} = 0, \quad V^{C_m} = 0. \quad (8)$$

The relabeling α_m^{-1} gives the displayed outer index $W \subset \mathcal{U}^{C_m}$, and root pullbacks identify the outer fixed-point group with C_{mn}/C_m . The identity for the α_n ensures that both maps in (7) use the same final universe identification. For $L = \mathbb{R}^k \oplus W \oplus V$ one has $L^{C_m} = \mathbb{R}^k \oplus W$ and $L^{C_{mn}} = \mathbb{R}^k$. On (8), the left side of (7) first applies $r_{m,L}$ and then $r_{n,\rho_m^* L^{C_m}}$. The right side sends (W, V) to $W \oplus V$ and applies $r_{mn,L}$. These are equal by the strict identity in BM condition (iii). Both composites collapse all bar coordinates; the identity also commutes with every indexing map. Hence it holds on the whole double realization. The unit identity follows from $r_{1,L} = \text{id}$.

Each R_n is a stable equivalence on underlying spectra. Indeed, its map on stable homotopy groups is induced by the same $r_{n,L}$ as in BM condition (iv). The comparison with the geometric

stable homotopy groups is [13, Lemma II.2.11], which is natural in these maps. Now R_n , R_{mn} , and $\mu_{n,m}$ in (7) are underlying stable equivalences. It follows that $D_n(R_m)$ is an underlying stable equivalence for every n . Since finite geometric fixed points detect equivalences for the family of finite cyclic subgroups, each R_m is an equivalence for that family. Thus $(T, (R_n))$ satisfies exactly the definition of Cyc_{NS}^O .

A BM morphism commutes with every $r_{n,L}$ and consequently with the cocones defining R_n . Taking the same map of underlying spectra defines Ξ on morphisms. Identities and compositions are preserved strictly. $\square$

Proposition 3.3. *There is an ∞ -functor*

$$F_{\text{BM}} : \text{NS} \xrightarrow{N(\Xi \text{THH}_{\text{BM}})} N\text{Cyc}_{\text{NS}}^O \xrightarrow{L_{\text{cyc}}} \text{CycSp}^{\text{gen}}. \quad (9)$$

Proof. Blumberg–Mandell construct an ordinary functor $\text{THH}_{\text{BM}} : \mathcal{S} \rightarrow \text{Cyc}_{\text{BM}}$ in [6, Theorem 4.9]. Its structure maps restrict equivariant maps to fixed points after n -fold edgewise subdivision. A fixed cyclic word and its Bökstedt index are repeated n times, and the restriction retains one period. For a word repeated mn times, successively retaining periods for m and n agrees with retaining one period for mn . Restriction to nested fixed subspaces has the same strict compatibility. These operations commute with applying a strict enriched functor to the entries of the word.

Compose this ordinary functor with Ξ and take its nerve. The last arrow of (9) is the localization functor of [13, Theorem II.3.7]. Thus both arrows are defined on their entire source ∞ -categories, and their composite gives the asserted functor. $\square$

3.3 Morita invariance and descent

Lemma 3.4. *A morphism in Cyc_{NS}^O which is a stable equivalence on underlying spectra is an equivalence for the family of finite cyclic subgroups. It therefore becomes an equivalence under L_{cyc} .*

Proof. For such a morphism $f : T \rightarrow T'$, the square

$$\begin{array}{ccc} D_n T & \xrightarrow{D_n f} & D_n T' \\ R_n \downarrow & & \downarrow R'_n \\ T & \xrightarrow{f} & T' \end{array}$$

commutes. The vertical maps and the bottom map are underlying stable equivalences, so the top map is one as well. Apply detection by finite geometric fixed points. $\square$

Proposition 3.5. *The functor F_{BM} inverts every Morita equivalence.*

Proof. Let $f : \mathcal{A} \rightarrow \mathcal{B}$ be a Morita equivalence. The functorial pointwise fibrant replacement of [6, Proposition 2.6] gives a natural square from f to $f^\Omega : \mathcal{A}^\Omega \rightarrow \mathcal{B}^\Omega$, whose vertical maps are DK equivalences. They are sent to stable equivalences by [6, Theorem 5.9]. They also induce equivalences of perfect-module categories, so f^Ω remains a Morita equivalence. It suffices to consider f^Ω .

Representable modules over these pointwise fibrant spectral categories are cofibrant and fibrant in the projective module model structures. Derived extension of scalars sends the representable h_a to h_{fa} . Its full faithfulness on perfect modules and enriched Yoneda therefore show that

$$\mathcal{A}^\Omega(a, a') \longrightarrow \mathcal{B}^\Omega(fa, fa')$$

is a stable equivalence for all a, a' . Moreover, the representables h_{fa} thickly generate $\text{Perf}(\mathcal{B}^\Omega)$. Let $\mathcal{C}$ be the full spectral subcategory of $\mathcal{B}^\Omega$ on the image objects. Then $\mathcal{A}^\Omega \rightarrow \mathcal{C}$ is a DK equivalence.

Choose a small full spectral category $\mathcal{P}$ of cofibrant-fibrant perfect right $\mathcal{B}^\Omega$ -modules which contains all representables and represents every perfect-module homotopy class. Keeping object labels for the representables makes Yoneda a strict full inclusion $\mathcal{B}^\Omega \subset \mathcal{P}$. The category $\mathcal{P}$ is pretriangulated in the sense of [6, Definition 5.4]. To see the required representability, choose cofibrant-fibrant perfect representatives of the zero module, shifts, and homotopy cofibers. Mapping out of a cofibrant module preserves the corresponding homotopy cofiber sequences because the module category is stable. The chosen objects therefore represent the shifted and cofiber modules over $\mathcal{P}$. This is the perfect-module construction used in the proof of [6, Theorem 5.5].

In $\pi_0\mathcal{P}$, the objects of $\mathcal{B}^\Omega$ have thick closure all of $\pi_0\mathcal{P}$, by the definition of a perfect module. The objects of $\mathcal{C}$ have the same thick closure, by the Morita hypothesis. Thus [6, Theorem 5.12] applies to both full inclusions

$$\mathcal{C} \subset \mathcal{P}, \quad \mathcal{B}^\Omega \subset \mathcal{P}.$$

Both induced THH maps are stable equivalences. Their commuting triangle implies that $\text{THH}_{\text{BM}}(\mathcal{C}) \rightarrow \text{THH}_{\text{BM}}(\mathcal{B}^\Omega)$ is a stable equivalence. DK invariance for $\mathcal{A}^\Omega \rightarrow \mathcal{C}$, followed by the replacement square, proves that $\text{THH}_{\text{BM}}(f)$ is a stable equivalence. It is already a cyclotomic morphism, so Lemma 3.4 finishes the proof. $\square$

The auxiliary category $\mathcal{P}$ tests the equivalence of a given map. The functor in (9) was defined before this choice and does not require a functorial choice of $\mathcal{P}$.

The localization (5) induces an equivalence of functor ∞ -categories

$$q^* : \text{Fun}(\text{Cat}^{\text{perf}}, \text{CycSp}^{\text{gen}}) \xrightarrow{\simeq} \text{Fun}_{W_{\text{Mor}}}(\text{NS}, \text{CycSp}^{\text{gen}}), \quad (10)$$

where the right side is the full subcategory of functors inverting W_{Mor} . By Proposition 3.5, F_{BM} is an object of this subcategory. Define THH^{gen} by lifting that object through (10), together with a natural equivalence

$$\eta : q^*\text{THH}^{\text{gen}} \xrightarrow{\simeq} F_{\text{BM}}. \quad (11)$$

This construction takes place in the functor ∞ -category, so it supplies the functor on morphisms and all higher simplices at once.

Proposition 3.6. *The space of pairs*

$$(F, \eta), \quad F : \text{Cat}^{\text{perf}} \rightarrow \text{CycSp}^{\text{gen}}, \quad \eta : q^*F \simeq F_{\text{BM}},$$

is contractible.

Proof. Take maximal subgroupoids in (10). The result is an equivalence of spaces, and the space in the statement is its homotopy fiber over F_{BM} . $\square$

The specified comparison with F_{BM} is part of this uniqueness statement. It does not assert uniqueness of all genuine refinements of an arbitrary NS cyclotomic spectrum.

3.4 Good models for the cyclic comparison

We next compare the functors on NS before descending the comparison. We require a model with cofibrant units in order to control cyclic realization.

Apply geometric realization to the mapping spectra of $\mathcal{A}$. This is strong symmetric monoidal and takes cofibrant simplicial symmetric spectra to cofibrant topological symmetric spectra [10, Section 19, Theorem 19.3]. On topological spectral categories use the functorial replacement

$$Q|\mathcal{A}| \longrightarrow |\mathcal{A}|$$

of [7, Definition 4.5 and Proposition 4.6]. It preserves the object set and is a natural DK equivalence. Its off-diagonal mapping spectra are cofibrant, and each unit $\mathbb{S} \rightarrow Q|\mathcal{A}|(x, x)$ is a cofibration. Hence all mapping spectra are cofibrant, and the units on zeroth spaces are closed h -cofibrations.

Let P be prolongation from topological symmetric spectra to orthogonal spectra and R its restriction functor. Prolongation is strong symmetric monoidal, its adjunction unit is monoidal, and the underived unit $X \rightarrow RPX$ is a stable equivalence for cofibrant X ; see [10, Propositions 3.3–3.4, Lemmas 10.2–10.3, and Theorem 10.4]. Consequently

$$|\mathcal{A}| \longleftarrow Q|\mathcal{A}| \longrightarrow RPQ|\mathcal{A}| \quad (12)$$

is a natural zigzag of strict enriched DK equivalences. Put $\mathcal{A}^O = PQ|\mathcal{A}|$. Its mapping spectra and units satisfy the cofibrancy conditions just stated.

The same Bökstedt formula defines the topological version of $\mathrm{THH}_{\mathrm{BM}}$; the DK-invariance proof of [6, Theorem 5.9] applies to it. Thus (12) induces natural cyclotomic equivalences for that construction. On the cyclic-bar side, the zigzag preserves object sets and is an equivalence on each mapping spectrum. Derived tensor products, coproducts of words, and realization preserve these equivalences. Naturality of the Tate diagonal gives the compatibility with the prime Frobenius maps. This comparison uses only equivalences on individual mapping spectra. It proves the required model invariance without using Morita invariance of the cyclic-bar construction.

It therefore suffices to construct the comparison for the good orthogonal models $\mathcal{A}^O$. The construction of these models is natural for all morphisms in $\mathbb{S}$, including functors whose object functions are not injective.

3.5 The total category of cyclic words

For an object set O , the assignment $[r] \mapsto O^{r+1}$ is a cyclic set: faces delete vertices, degeneracies repeat vertices, and the cyclic operator rotates them. Its Grothendieck construction is a discrete opfibration

$$\pi : \mathrm{Word}_O \longrightarrow \Lambda^{\mathrm{op}}.$$

For a word $w = (x_0, \dots, x_r)$ define the tuple and its derived tensor product by

$$X_{\mathcal{A}}(w) = (\mathcal{A}^O(x_0, x_1), \dots, \mathcal{A}^O(x_r, x_0)), \quad M_{\mathcal{A}, w} = \bigotimes X_{\mathcal{A}}(w).$$

These tuples define a functor to the active symmetric monoidal category of orthogonal spectra. A face uses permutation and composition of adjacent mapping spectra; a degeneracy inserts a unit; a cyclic operator permutes the factors. Empty-input operations are the units. The enriched associativity and unit identities give all relations. In particular, the last face is the composite of a cyclic permutation and an adjacent composition within this same functor.

To keep the dependence on $\mathcal{A}$, form the ordinary Grothendieck construction of $(\mathcal{A}, [r]) \mapsto \mathrm{Ob}(\mathcal{A})^{r+1}$:

$$\Pi : \mathrm{Word} = \int_{\mathbb{S} \times \Lambda^{\mathrm{op}}} \mathrm{Ob}(\mathcal{A})^{r+1} \longrightarrow \mathbb{S} \times \Lambda^{\mathrm{op}}. \quad (13)$$

The tuple construction is a functor on this whole ordinary category. Its morphisms include both cyclic operations and strict enriched functors. Hence a natural transformation pulled back to $N\mathrm{Word}$ already includes compatibility with their compositions and higher simplices.

Let B be the Bökstedt tuple functor of [13, Definition III.4.3]. Define

$$C_r(\mathcal{A}) = \bigvee_{w \in \mathrm{Ob}(\mathcal{A})^{r+1}} B(X_{\mathcal{A}}(w)), \quad C(\mathcal{A}) = |C_{\bullet}(\mathcal{A})|.$$

The tensor-product comparison of [13, Theorem III.4.5] is a natural equivalence of lax symmetric monoidal ∞ -functors. Precompose it with the tuple functor on $N\mathbf{Word}$, and left Kan extend along $N\Pi$. Because Π is a discrete opfibration, the value of this Kan extension at $(\mathcal{A}, [r])$ is the coproduct over the word fiber. We obtain a natural equivalence in

$$\mathrm{Fun}(NS \times N\Lambda^{\mathrm{op}}, \mathrm{Sp}).$$

Currying and applying cyclic realization produces

$$e : \mathrm{THH}_{\mathrm{cyc}} \xrightarrow{\cong} C \quad \text{in } \mathrm{Fun}(NS, \mathrm{Sp}^{B\mathbb{T}}), \quad (14)$$

where the derived cyclic bar $\mathrm{THH}_{\mathrm{cyc}}$ has degree- r term $\bigoplus_w M_{\mathcal{A}, w}$.

3.6 The comparison with the classical Bökstedt construction

For a representation V , the classical degree- r formula is

$$T_r(\mathcal{A})(V) = \mathrm{hocolim}_{\mathbf{m} \in \mathcal{I}^{r+1}} \mathrm{Map}_*(S^{|\mathbf{m}|}, S^V \wedge \bigvee_w Y_{w, \mathbf{m}}),$$

where $\mathcal{I}$ is the category of finite sets and injections, $|\mathbf{m}| = m_0 + \cdots + m_r$, and $Y_{w, \mathbf{m}}$ is the smash product of the corresponding mapping-spectrum spaces. The word inclusions define actual maps

$$\bigvee_w \mathrm{Map}_*(S^{|\mathbf{m}|}, S^V \wedge Y_{w, \mathbf{m}}) \longrightarrow \mathrm{Map}_*(S^{|\mathbf{m}|}, S^V \wedge \bigvee_w Y_{w, \mathbf{m}}). \quad (15)$$

They commute with index injections, block concatenations, block permutations, and maps of mapping spectra. After taking homotopy colimits they give a strict cyclic natural map

$$\kappa_\bullet : C_\bullet(\mathcal{A}) \longrightarrow T_\bullet(\mathcal{A}).$$

The last face commutes with this map because both its block permutation and its enriched composition commute with the word inclusions in (15).

For fixed $\mathbf{m}$, regard (15) as a map of orthogonal spectra in V . Its two sides represent

$$\bigoplus_w \Sigma^{-|\mathbf{m}|} \Sigma^\infty Y_{w, \mathbf{m}} \quad \text{and} \quad \Sigma^{-|\mathbf{m}|} \Sigma^\infty \left(\bigvee_w Y_{w, \mathbf{m}} \right).$$

Suspension spectra and finite desuspension preserve coproducts, so this map is a stable equivalence for arbitrary word sets. The good-model conditions ensure that the Bousfield–Kan formulas compute derived homotopy colimits [13, Appendix C, Proposition C.11].

The cyclic diagrams are proper. A degeneracy repeats a vertex x and inserts the Bökstedt index 0, with the closed h -cofibration $S^0 \rightarrow \mathcal{A}^O(x, x)_0$ in that factor. In bar degree q , the index insertion selects precisely the chains $\mathbf{m}^0 \rightarrow \cdots \rightarrow \mathbf{m}^q$ whose inserted coordinate is 0 at every object; every arrow in that coordinate is then id_0 . Together with the repeated-vertex word labels and the unit subspaces in their coefficient spaces, these chains form a simplicial subspace of the Bousfield–Kan bar construction. Well-pointedness of the other factors makes the unit inclusions and the inclusions of the selected word subwedges closed h -cofibrations. Mapping out of the compact sphere $S^{|\mathbf{m}|}$ preserves these inclusions as closed h -cofibrations. This applies both to the individual word terms of $C_\bullet$ and to the word wedges inside the mapping spaces of $T_\bullet$. Filter the two bar constructions by their skeleta. For a nondegenerate indexing chain σ of length q , write $A_\sigma \rightarrow B_\sigma$ for its coefficient inclusion, taking $A_\sigma = *$ when the chain is absent from the source. Its relative attaching map is

$$(A_\sigma \wedge \Delta_+^q) \cup_{A_\sigma \wedge \partial \Delta_+^q} (B_\sigma \wedge \partial \Delta_+^q) \longrightarrow B_\sigma \wedge \Delta_+^q.$$

This is the pushout-product of a closed h -cofibration with the boundary inclusion. Coproducts, pushouts, and successive skeletal attachments preserve these cofibrations, so the induced degeneracy maps are h -cofibrations. Thus the argument of [13, Lemma III.5.2], followed by the Lewis criterion used there, makes the latching maps h -cofibrations. Realization consequently preserves the degreewise stable equivalences and gives

$$\kappa : C \xrightarrow{\simeq} \mathrm{THH}_{\mathrm{BM}} \quad \text{in } \mathrm{Fun}(\mathcal{NS}, \mathrm{Sp}^{B\mathbb{T}}). \quad (16)$$

3.7 Repeated words and the residual circle action

Fix a prime p . Let $q_p : \Lambda_p \rightarrow \Lambda$ and $\mathrm{sd}_p : \Lambda_p \rightarrow \Lambda$ be the quotient and subdivision functors. Pull Word_O back along q_p^{op} . Repetition sends a word w to the long word w^p and defines a functor from this pullback to the pullback along $\mathrm{sd}_p^{\mathrm{op}}$. The edge positions of w^p form a free C_p -set with quotient the edge positions of w , each labelled by its mapping spectrum. Before localization, repetition gives an ordinary BC_p -equivariant tuple functor

$$\mathrm{Word}_{O,p} \longrightarrow \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)_{\mathrm{act}}^{\otimes}.$$

Taking its nerve and localizing the orthogonal spectra gives

$$N\mathrm{Word}_{O,p} \longrightarrow N(\mathrm{Free}_{C_p}) \times_{N(\mathrm{Fin})} \mathrm{Sp}_{\mathrm{act}}^{\otimes}. \quad (17)$$

Here Free_{C_p} is the category of finite free C_p -sets, the functor to Fin takes the quotient, and the equivariance is rotation of the repeated blocks. The same construction for all spectral categories is indexed by

$$\Pi_p : \mathrm{Word}_p = \mathrm{Word} \times_{\Lambda^{\mathrm{op}}} \Lambda_p^{\mathrm{op}} \longrightarrow \mathcal{S} \times \Lambda_p^{\mathrm{op}}.$$

All the transformations below are performed on this total diagram.

The geometric-fixed-point tuple functor $B_p^{\mathbb{T}}$ of [13, Section III.5] retains the circle action on the universe representations. Denote this circle by $\mathbb{T}_{\mathrm{univ}}$ and the circle obtained from cyclic realization by $\mathbb{T}_{\mathrm{cyc}}$. First take the BC_p -equivariant left Kan extension along $N\Pi_p$ with values in $\mathrm{Sp}^{B\mathbb{T}_{\mathrm{univ}}}$. Then apply the realization functor of [13, Propositions B.21–B.22]. The two actions descend to $(\mathbb{T}_{\mathrm{cyc}} \times \mathbb{T}_{\mathrm{univ}})/C_p$; restriction along the diagonal $\mathbb{T}/C_p$ and pullback by ρ_p give the required circle action. For tuple functors with trivial universe action which descend to Λ^{op} , this construction is ordinary cyclic realization by those same propositions.

We identify the fixed-point vertex explicitly. In degree r of $\mathrm{sd}_p C_{\bullet}$, the group C_p rotates both the word labels and the corresponding blocks of spectra. The fixed labels are precisely w^p . Each remaining orbit is free, so its summands form an induced C_p -spectrum and have zero geometric fixed points. The fixed-label inclusions therefore induce an equivalence

$$\Phi_{\mathcal{U}}^{C_p}((\mathrm{sd}_p C)_r) \simeq \bigoplus_{w \in O^{r+1}} B_p^{\mathbb{T}}(X_{\mathcal{A}}(w)). \quad (18)$$

On the left, geometric fixed points are applied to the degreewise C_p -action, with the additional universe-circle action retained. The equality is an equivalence of the full equivariant Λ_p^{op} -diagrams. It is also natural for noninjective object functions: if a nonperiodic word maps to a periodic word, its source geometric-fixed-point contribution is still zero.

Subdivision realizes equivariantly to the original cyclic realization [13, Proposition B.19]. Properness permits finite fixed points to commute with this realization; the universe homotopy colimit in (6) commutes with realization as well. Apply these interchanges at each representation and then (18). The diagonal realization of its right side is thereby naturally identified with $D_p C(\mathcal{A})$. Throughout this calculation the action on each sphere S^V is retained. Thus the residual action is the diagonal action of [13, Propositions B.21–B.22]. This is the labelled form of the argument in [13, Propositions III.5.5–III.5.6].

Pull back the restriction map $B_p^\mathbb{T} \rightarrow B$ of [13, Construction III.4.6 and Theorem III.4.7] along the ordinary repeated-word functor preceding (17), take the equivariant word Kan extension, and perform this diagonal realization. The result is a natural equivalence

$$c_p : D_p C \xrightarrow{\cong} C. \quad (19)$$

Its equivalence follows from [13, Theorem III.4.7] on every fixed word, followed by coproducts and realization.

The map κ commutes with these restrictions. On a repeated word with diagonal Bökstedt index, both composites restrict an equivariant based map to fixed points and include the corresponding word summand. Nonperiodic source orbits contribute zero geometric fixed points. Indexing maps, root pullbacks, and the universe-circle actions are preserved. Hence the square

$$\begin{array}{ccc} D_p C & \xrightarrow{D_p \kappa} & D_p \mathrm{THH}_{\mathrm{BM}} \\ c_p \downarrow & & \downarrow R_p \\ C & \xrightarrow{\kappa} & \mathrm{THH}_{\mathrm{BM}} \end{array} \quad (20)$$

commutes naturally. Invert (19) in the functor ∞ -category and compose with the natural geometric-fixed-point-to-Tate map. This gives C its NS Frobenius maps. Square (20) makes κ a morphism of these cyclotomic functors.

3.8 The natural Frobenius comparison

It remains to prove that e in (14) respects the Tate-diagonal Frobenius. The universal comparison in [13, Theorem III.6.7 and the following comparison diagram] involves the tensor product I of the quotient tuple, B , $B_{p,\mathrm{lax}}^\mathbb{T}$, $B_{p,\infty}$, and the Tate construction $\tilde{T}_p$ of the repeated tuple. These functors descend to the target of (17), with values in spectra carrying the universe-circle action. The map $I \rightarrow B$ is the tensor comparison; $I \rightarrow \tilde{T}_p$ is the multivariable Tate diagonal of [13, Corollary III.3.8]. The initiality statement of [13, Theorem III.6.7] gives the homotopy comparing the route through $B_{p,\mathrm{lax}}^\mathbb{T}$ and $B_{p,\infty}$ with this Tate diagonal.

The point-set functor $B_p^\mathbb{T}$ is not included in that lax symmetric monoidal descent statement. To include it, pull the preceding diagram back along the localization of the point-set tuple category and use the ordinary repeated-word functor defined before (17). The natural transformation $B_p^\mathbb{T} \rightarrow B_{p,\mathrm{lax}}^\mathbb{T}$ and its triangle with B commute on that point-set category, as in the comparison diagram following [13, Theorem III.6.7]. Its other composite, through $B_{p,\infty}$, is the geometric-fixed-point-to-Tate map. Thus all vertices, including $B_p^\mathbb{T}$, and the comparison homotopy are defined on a single common domain. The resulting diagram takes values in $\mathrm{Sp}^{B\mathbb{T}_{\mathrm{univ}}}$ and retains its BC_p -equivariance.

Precompose the whole diagram with the ordinary repeated-word functor, left Kan extend over the word fibers, and apply diagonal realization. At the Tate vertices use the natural maps

$$\bigoplus_w (M_{\mathcal{A},w}^{\otimes p})^{tC_p} \longrightarrow \left(\bigoplus_{\text{long words } v} M_{\mathcal{A},v} \right)^{tC_p}, \quad (21)$$

$$|Y_\bullet|^{tC_p} \longrightarrow |Y_\bullet|^{tC_p}. \quad (22)$$

The first map is induced by the inclusions of repeated-word summands, and the second by the realization cocone. Compatibility of (22) with residual circle actions is [13, Proposition B.20]. These constructions use these maps in their displayed direction; no interchange of Tate with a coproduct or a realization is needed.

Because these operations act on the full universal diagram and its homotopy, they give a specified natural homotopy making

$$\begin{array}{ccc}
\mathrm{THH}_{\mathrm{cyc}} & \xrightarrow{e} & C \\
\varphi_p^{\mathrm{NS}} \downarrow & & \downarrow \varphi_p^C \\
\rho_p^* \mathrm{THH}_{\mathrm{cyc}}^{tC_p} & \xrightarrow{\rho_p^* e^{tC_p}} & \rho_p^* C^{tC_p}
\end{array} \tag{23}$$

commute in $\mathrm{Fun}(\mathrm{NS}, \mathrm{Sp}^{B\mathbb{T}})$. The root pullback here relabels the residual action after Tate; the root pullback in D_p already belongs to the geometric-fixed-point construction.

The lax-equalizer definition of $\mathrm{CycSp}_{\mathrm{NS}}$ packages a circle spectrum and its prime Frobenius maps, and packages a morphism together with the corresponding homotopies. Since $\mathrm{Fun}(\mathrm{NS}, -)$ preserves limits, the natural squares (23) for all primes define a morphism in $\mathrm{Fun}(\mathrm{NS}, \mathrm{CycSp}_{\mathrm{NS}})$. Its underlying circle-spectrum map is the equivalence e . The same is true for κ . Conservativity of the underlying-spectrum functor [13, Corollary II.1.7] therefore shows that both are cyclotomic equivalences.

Including the natural model comparisons (12), we have constructed an actual cyclic-word functor and an equivalence

$$H_{\mathrm{NS}} : \mathrm{NS} \longrightarrow \mathrm{CycSp}_{\mathrm{NS}}, \quad H_{\mathrm{NS}} \xrightarrow{\simeq} UF_{\mathrm{BM}} \quad \text{in } \mathrm{Fun}(\mathrm{NS}, \mathrm{CycSp}_{\mathrm{NS}}). \tag{24}$$

By Proposition 3.5, this functor inverts W_{Mor} . It consequently descends along q ; its descent is $\mathrm{THH}_{\mathrm{NS}}$ with its cyclic-bar definition. If $\mathcal{A}$ is pretriangulated and its homotopy category is idempotent complete, Yoneda identifies $q(\mathcal{A})$ with the stable ∞ -category presented by $\mathcal{A}$ after fibrant replacement. The value of the descended functor is then the derived cyclic bar of its mapping spectra, with the multivariable Tate-diagonal Frobenius just constructed. In particular, the comparison has not assumed in advance the THH equivalence from $\mathcal{A}$ to $\mathrm{Perf}(\mathcal{A})$.

Proof of Theorem 3.1. The functor $\mathrm{THH}^{\mathrm{gen}}$ and its specified source comparison were constructed in (10)–(11). Combining (24) with that comparison gives

$$q^* \mathrm{THH}_{\mathrm{NS}} \simeq q^*(U\mathrm{THH}^{\mathrm{gen}}) \quad \text{in } \mathrm{Fun}(\mathrm{NS}, \mathrm{CycSp}_{\mathrm{NS}}).$$

The Morita-localization universal property for target $\mathrm{CycSp}_{\mathrm{NS}}$ makes q^* fully faithful. Thus the displayed equivalence descends to the natural equivalence in the theorem. $\square$

4 Realization of bounded-below cyclotomic spectra

Theorem 4.1. *Let $X \in \mathrm{CycSp}_{\mathrm{NS}}$ have bounded-below underlying spectrum. There are a small idempotent-complete stable ∞ -category $\mathcal{C}$ and an equivalence*

$$\mathrm{THH}(\mathcal{C}) \simeq X \quad \text{in } \mathrm{CycSp}_{\mathrm{NS}}.$$

If $E \in \mathrm{CycSp}^{\mathrm{gen}}$ has bounded-below underlying spectrum, there is likewise such a category $\mathcal{C}$ with $\mathrm{THH}^{\mathrm{gen}}(\mathcal{C}) \simeq E$.

We lift a cellular approximation of X to localizing motives. This requires a comparison of mapping spectra at each finite stage and enough cell maps to make the approximation converge. Efimov’s comparison for connective ring spectra supplies the first requirement after an arbitrary coproduct is exhibited as a retract of one ring motive. The corepresentability of integral TR supplies the second requirement by giving a surjection onto the zeroth homotopy group of every connective target. Full faithfulness at the finite stages then lifts the entire sequential diagram, including its coherent structure maps.

4.1 Reduced polynomial motives and the comparison maps

Cyclotomic-valued THH is a finitary localizing invariant; this is the form of THH used in [8, §9.1, immediately before (9.4)], and its localizing property is also used in [12, proof of Corollary 2.4.7]. The filtered-colimit and localization comparisons can be detected after applying V , where they are the corresponding comparisons for ordinary THH [4, Section 10.1], by Lemma 2.1. The universal property of localizing motives [15, Theorem 1.5(3)] therefore gives a colimit-preserving exact functor

$$\mathrm{THH}_{\mathrm{mot}} : \mathrm{Mot}_{\mathrm{loc}} \longrightarrow \mathrm{CycSp}_{\mathrm{NS}}, \quad \mathrm{THH}_{\mathrm{mot}} U_{\mathrm{loc}}(\mathcal{C}) \simeq \mathrm{THH}(\mathcal{C}). \quad (25)$$

For an $\mathbb{E}_1$ -ring spectrum R , write $U_{\mathrm{loc}}(R) = U_{\mathrm{loc}}(\mathrm{Perf}(R))$.

The unit of $\mathbb{S}[t] = \Sigma_+^\infty \mathbb{N}$ splits the augmentation $t \mapsto 0$. Define

$$\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t]) := \mathrm{fib}(U_{\mathrm{loc}}(\mathbb{S}[t]) \longrightarrow U_{\mathrm{loc}}(\mathbb{S})), \quad (26)$$

$$\widetilde{\mathrm{THH}}(\mathbb{S}[t]) := \mathrm{fib}(\mathrm{THH}(\mathbb{S}[t]) \longrightarrow \mathrm{THH}(\mathbb{S})). \quad (27)$$

Both fibers are retracts of their respective polynomial objects. Exactness in (25) gives

$$\mathrm{THH}_{\mathrm{mot}}(\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])) \simeq \widetilde{\mathrm{THH}}(\mathbb{S}[t]).$$

The reduced THH object has connective underlying spectrum: $\mathbb{S}[t]$ is connective, its cyclic bar construction is degreewise connective, and the reduced object is a retract of its realization. The splitting also identifies (26) with the cofiber of the unit map used in Efimov's definition of the reduced motive.

We use two comparison results in their natural mapping-spectrum form. For every connective $\mathbb{E}_1$ -ring spectrum R , the map induced by the actual functor $\mathrm{THH}_{\mathrm{mot}}$ is an equivalence

$$\begin{aligned} \mathrm{map}_{\mathrm{Mot}_{\mathrm{loc}}}(\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t]), U_{\mathrm{loc}}(R)) \\ \xrightarrow{\simeq} \mathrm{map}_{\mathrm{CycSp}_{\mathrm{NS}}}(\widetilde{\mathrm{THH}}(\mathbb{S}[t]), \mathrm{THH}(R)). \end{aligned} \quad (28)$$

This is Efimov's comparison in [8, Theorem 9.1 and (9.4)]; the case $k = 1$ of [8, Proposition 9.4 and its proof] also identifies the comparison with the one induced by THH.¹ In particular, (28) specifies which map is an equivalence, rather than just identifying its source and target abstractly with $\mathrm{TR}(R)$.

For every bounded-below $X \in \mathrm{CycSp}_{\mathrm{NS}}$, McCandless's corepresentability theorem [12, Theorem 3.3.12] gives a natural equivalence

$$\mathrm{map}_{\mathrm{CycSp}_{\mathrm{NS}}}(\widetilde{\mathrm{THH}}(\mathbb{S}[t]), X) \simeq \mathrm{TR}^{\mathrm{gen}}(E) = \lim_{n, R} E^{C_n}, \quad (29)$$

where E is the genuine object corresponding to X under (4). The limit is over the positive integers with an arrow $m \rightarrow n$ when $n \mid m$. Its transition maps are the restriction maps, constructed from categorical-to-geometric fixed points and the genuine cyclotomic equivalences [12, Construction 3.3.5 and Definition 3.3.6].

4.2 Coproducts as retracts of a connective ring motive

For an object Z in a category with small coproducts, write $Z^{(I)} = \bigoplus_{i \in I} Z$.

¹In the displayed formula (9.4) of arXiv:2510.17010v1, the second variable on the motivic side is printed as $U_{\mathrm{loc}}(\mathbb{S})$. We use $U_{\mathrm{loc}}(R)$, as specified by the preceding statement for connective $\mathbb{E}_1$ -rings, Theorem 9.1, and the proof of Proposition 9.4, which retains the factor R in the relative cyclotomic-trace comparison.

Lemma 4.2. *For every small set I there is a connective $\mathbb{E}_1$ -ring spectrum R_I with an equivalence*

$$U_{\text{loc}}(R_I) \simeq U_{\text{loc}}(\mathbb{S}) \oplus \bigoplus_{i \in I} U_{\text{loc}}(\mathbb{S}[t]). \quad (30)$$

Consequently, $\widetilde{U}_{\text{loc}}(\mathbb{S}[t])^{(I)}$ is a retract of $U_{\text{loc}}(R_I)$.

Proof. Let $\text{Fin}(I)$ be the filtered poset of finite subsets of I . For $J \in \text{Fin}(I)$ put

$$R_J = \mathbb{S} \times \prod_{j \in J} \mathbb{S}[t],$$

with augmentation $\epsilon_J : R_J \rightarrow \mathbb{S}$ given by the first projection. If $J \subseteq K$, use the unital map

$$(s, (b_j)_{j \in J}) \mapsto (s, (b_j)_{j \in J}, (s \cdot 1)_{k \in K \setminus J}). \quad (31)$$

Each new coordinate is the composite $R_J \xrightarrow{\epsilon_J} \mathbb{S} \rightarrow \mathbb{S}[t]$. For $J \subseteq K \subseteq L$, a coordinate indexed by J is unchanged by both transitions, and every coordinate indexed by $L \setminus J$ is the image of the same first coordinate under the unit. Thus the composite is the map prescribed for $J \subseteq L$. These formulas specify an actual diagram of $\mathbb{E}_1$ -rings: one may represent the arrow $\mathbb{S} \rightarrow \mathbb{S}[t]$ by an objectwise fibrant arrow of associative ring spectra and form its strict finite products. The formulas then compose strictly, and these products compute the products of $\mathbb{E}_1$ -rings.

The maps ϵ_J and the diagonal sections

$$\delta_J : \mathbb{S} \longrightarrow R_J, \quad s \longmapsto (s, (s \cdot 1)_{j \in J})$$

are natural in J , and $\epsilon_J \delta_J = \text{id}$. The finite-product decomposition of module categories gives an exact sequence

$$\mathcal{K}_J \longrightarrow \text{Perf}(R_J) \xrightarrow{- \otimes_{R_J} \mathbb{S}} \text{Perf}(\mathbb{S}), \quad \mathcal{K}_J \simeq \bigoplus_{j \in J} \text{Perf}(\mathbb{S}[t]). \quad (32)$$

The functors between the kernels in this diagram are the standard inclusions of summands. Indeed, if $e_j \in \pi_0 R_J$ is the idempotent of the j th polynomial coordinate, then its first coordinate is zero, so (31) sends it to the corresponding $e_j \in \pi_0 R_K$. Moreover, the R_J -action on $e_j R_K$ still factors through the same polynomial coordinate. Extension of scalars is therefore the identity on the j th $\text{Perf}(\mathbb{S}[t])$ and has zero component in every new summand.

Applying U_{loc} to (32) gives a natural cofiber sequence

$$\bigoplus_{j \in J} U_{\text{loc}}(\mathbb{S}[t]) \longrightarrow U_{\text{loc}}(R_J) \longrightarrow U_{\text{loc}}(\mathbb{S}).$$

The natural section induced by δ_J splits this sequence in the stable diagram category $\text{Fun}(N\text{Fin}(I), \text{Mot}_{\text{loc}})$. In particular, its splitting is compatible with the transitions in J .

Set $R_I = \text{colim}_{J \in \text{Fin}(I)} R_J$. Filtered colimits of $\mathbb{E}_1$ -rings are computed on underlying spectra, so R_I is connective. The functor $R \mapsto \text{Perf}(R)$ preserves filtered colimits with values in Cat^{perf} : finite cell modules and the finite diagrams defining them descend to a stage, mapping spectra between these modules commute with the filtered colimit, and taking retracts gives the idempotent-complete statement. Since U_{loc} preserves filtered colimits, the compatible splitting above gives

$$\begin{aligned} U_{\text{loc}}(R_I) &\simeq \text{colim}_{J \in \text{Fin}(I)} U_{\text{loc}}(R_J) \\ &\simeq U_{\text{loc}}(\mathbb{S}) \oplus \bigoplus_{i \in I} U_{\text{loc}}(\mathbb{S}[t]). \end{aligned}$$

Finally, take the coproduct of the retractions of $U_{\text{loc}}(\mathbb{S}[t])$ onto its reduced summand and compose it with (30). $\square$

Corollary 4.3. *For all small sets I, J , the functor $\mathrm{THH}_{\mathrm{mot}}$ induces an equivalence*

$$\begin{aligned} \mathrm{map}_{\mathrm{Mot}_{\mathrm{loc}}}(\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])^{(I)}, \widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])^{(J)}) \\ \xrightarrow{\simeq} \mathrm{map}_{\mathrm{CycSp}_{\mathrm{NS}}}(\widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I)}, \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(J)}). \end{aligned} \quad (33)$$

Proof. First let I be a singleton. Apply (28) to the connective ring R_J constructed for the small indexing set J in Lemma 4.2, using its filtered-colimit construction when J is infinite. Restrict in the target variable to its retract $\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])^{(J)}$. The functor $\mathrm{THH}_{\mathrm{mot}}$ preserves this retraction and identifies its image with $\widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(J)}$; hence the restricted comparison is an equivalence. For arbitrary I , mapping out of a coproduct gives the product, indexed by I , of these equivalences. $\square$

4.3 Full faithfulness on finite stable constructions

Let $\mathrm{Cell}_{\mathrm{mot}} \subseteq \mathrm{Mot}_{\mathrm{loc}}$ be the smallest full stable idempotent-complete subcategory containing $\widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])^{(I)}$ for every small set I . Define $\mathrm{Cell}_{\mathrm{cyc}} \subseteq \mathrm{CycSp}_{\mathrm{NS}}$ in the same way using $\widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I)}$. Each object in these subcategories is obtained from the specified coproducts by finitely many shifts, cofibers, and retracts. Every object of $\mathrm{Cell}_{\mathrm{cyc}}$ has bounded-below underlying spectrum, since its generators are connective and only finitely many stable constructions are performed.

Proposition 4.4. *The restriction of $\mathrm{THH}_{\mathrm{mot}}$ is an equivalence of ∞ -categories*

$$\mathrm{THH}_{\mathrm{mot}} : \mathrm{Cell}_{\mathrm{mot}} \xrightarrow{\simeq} \mathrm{Cell}_{\mathrm{cyc}}. \quad (34)$$

Proof. Fix a generator $A = \widetilde{U}_{\mathrm{loc}}(\mathbb{S}[t])^{(I)}$. The objects Z for which the comparison

$$\mathrm{map}_{\mathrm{Mot}_{\mathrm{loc}}}(A, Z) \longrightarrow \mathrm{map}_{\mathrm{CycSp}_{\mathrm{NS}}}(\mathrm{THH}_{\mathrm{mot}} A, \mathrm{THH}_{\mathrm{mot}} Z)$$

is an equivalence form a full subcategory closed under finite stable constructions and retracts. Indeed, both sides are exact in Z and preserve retract diagrams. By Corollary 4.3, this subcategory contains every generator, and consequently all of $\mathrm{Cell}_{\mathrm{mot}}$.

Now fix $Z \in \mathrm{Cell}_{\mathrm{mot}}$ and vary A . Mapping spectra take cofiber sequences in the source to fiber sequences, and the same argument shows that the comparison is an equivalence for every $A \in \mathrm{Cell}_{\mathrm{mot}}$. Thus the restriction of $\mathrm{THH}_{\mathrm{mot}}$ is fully faithful.

Its essential image contains the specified generators and is stable: full faithfulness lifts a morphism between image objects, and exactness carries its cofiber to the desired cofiber. It is also idempotent complete. A coherent idempotent diagram in the image lifts by full faithfulness, and its splitting exists in $\mathrm{Cell}_{\mathrm{mot}}$ and is preserved by the functor. The image therefore contains $\mathrm{Cell}_{\mathrm{cyc}}$. Conversely, every finite stable construction and retract defining an object of $\mathrm{Cell}_{\mathrm{mot}}$ maps into $\mathrm{Cell}_{\mathrm{cyc}}$. This proves essential surjectivity onto precisely the stated target. $\square$

4.4 Restriction maps and connective cell covers

The next lemma is the isotropy-separation connectivity argument underlying [3, Lemma 2.20 and Remark 2.23]; we include the relative form needed for restriction maps.

Lemma 4.5. *Let G be a finite group and let E be a genuine G -spectrum. Suppose that $\Phi^H E$ has connective underlying spectrum for every $H \leq G$. Then E^H is connective for every $H \leq G$. For every normal subgroup $N \trianglelefteq G$, the map*

$$E^G \longrightarrow (\Phi^N E)^{G/N}$$

has connective fiber, and hence is surjective on π_0 .

Proof. We first prove the assertion about E^H by induction on $|H|$. For $H = e$ it is the hypothesis. Let $\mathcal{P}_H$ denote the family of proper subgroups of H . Isotropy separation gives

$$(E\mathcal{P}_{H,+} \wedge E)^H \longrightarrow E^H \longrightarrow \Phi^H E. \quad (35)$$

The left term is computed from the nonnegative-dimensional H -CW filtration of $E\mathcal{P}_H$. For $K < H$, the projection formula and induction-coinduction equivalence for finite groups give

$$((H/K)_+ \wedge E)^H \simeq (\mathrm{Ind}_K^H \mathrm{Res}_K^H E)^H \simeq E^K.$$

Consequently, the fixed points of an r -dimensional relative orbit cell smashed with E are $\Sigma^r E^K$, with $r \geq 0$.

For a finite group, genuine categorical fixed points preserve all small colimits. More precisely, for $K \leq H$,

$$\pi_k(E^K) \cong [\Sigma^k \Sigma^\infty (H/K)_+, E]_{\mathrm{Sp}^H},$$

and the orbit spectrum representing this group is compact. Thus fixed points preserve filtered colimits. They are exact, and an exact functor preserving filtered colimits preserves arbitrary coproducts and hence all small colimits between stable categories. It follows that the left term of (35) is connective, by the induction hypothesis and the orbit-cell filtration. The right term is connective by assumption. Exactness then makes E^H connective.

For the relative assertion, use the family

$$\mathcal{F}_N = \{H \leq G \mid N \not\leq H\}.$$

Since N is normal, this is closed under conjugation and passage to subgroups. Relative isotropy separation gives a fiber sequence

$$(E\mathcal{F}_{N,+} \wedge E)^G \longrightarrow E^G \longrightarrow (\Phi^N E)^{G/N}. \quad (36)$$

Here the last term is identified with $(\widetilde{E}\mathcal{F}_N \wedge E)^G$ by the definition of relative geometric fixed points and iteration of categorical fixed points. The first term is again assembled from nonnegative suspensions of the spectra E^H , which have just been proved connective. It is therefore connective. Its π_{-1} vanishes, so the long exact sequence of (36) gives the claimed surjection. $\square$

Lemma 4.6. *For every $X \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq 0}$ there is a natural surjection*

$$\alpha_X : [\widetilde{\mathrm{THH}}(\mathbb{S}[t]), X]_{\mathrm{CycSp}_{\mathrm{NS}}} \twoheadrightarrow \pi_0(VX). \quad (37)$$

Consequently, there are a small set I and a map

$$a : \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I)} \longrightarrow X \quad (38)$$

in $\mathrm{CycSp}_{\mathrm{NS}}$ such that $\pi_0(Va)$ is surjective and $\mathrm{fib}(a) \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq 0}$.

Proof. Choose an inverse functor to the bounded-below comparison (4), and let E be the image of X . Every finite geometric fixed-point spectrum of E is connective: the genuine cyclotomic equivalences identify its underlying spectrum with VX . Apply Lemma 4.5 to the restriction of E to each finite cyclic group. All E^{C_n} are connective, and every restriction map

$$R_l : E^{C_{ln}} \longrightarrow (\rho_l^* \Phi^{C_l} E)^{C_n} \xrightarrow{\simeq} E^{C_n}$$

is surjective on π_0 .

Let Div have positive integers as objects and one arrow $a \rightarrow b$ when b divides a , so that $E^{C_a} \rightarrow E^{C_b}$ is a restriction map. The functor $\mathbb{N}_{\geq 1}^{\mathrm{op}} \rightarrow \mathrm{Div}$ sending n to $n!$ is initial: for every

m , the comma category of arrows $n! \rightarrow m$ is the nonempty tail of the reverse-ordered positive integers for which $m \mid n!$, and has contractible nerve. Thus [12, Remark 3.3.10] identifies

$$\mathrm{TR}^{\mathrm{gen}}(E) \simeq \lim_n (\cdots \rightarrow E^{C_{3!}} \rightarrow E^{C_{2!}} \rightarrow E^{C_1}).$$

Surjectivity of the transition maps on π_0 gives a surjection

$$\lim_n \pi_0(E^{C_{n!}}) \twoheadrightarrow \pi_0(VX) :$$

starting with a class in $\pi_0(VX)$, choose successive lifts along the tower. The Milnor exact sequence

$$0 \longrightarrow \lim_n^1 \pi_1(E^{C_{n!}}) \longrightarrow \pi_0 \mathrm{TR}^{\mathrm{gen}}(E) \longrightarrow \lim_n \pi_0(E^{C_{n!}}) \longrightarrow 0$$

therefore shows that the projection $\pi_0 \mathrm{TR}^{\mathrm{gen}}(E) \rightarrow \pi_0(VX)$ is surjective. Compose this projection with (29) to obtain (37). The inverse comparison functor, the corepresentability equivalence, and projection from the limit are natural, so the resulting α_X is natural in X .

To obtain actual cell maps on underlying homotopy groups, set

$$e = \alpha_{\widetilde{\mathrm{THH}}(\mathbb{S}[t])}(\mathrm{id}_{\widetilde{\mathrm{THH}}(\mathbb{S}[t])}) \in \pi_0(V\widetilde{\mathrm{THH}}(\mathbb{S}[t])).$$

This is defined because the reduced polynomial object is connective. For every $f : \widetilde{\mathrm{THH}}(\mathbb{S}[t]) \rightarrow X$, naturality gives

$$\alpha_X([f]) = (Vf)_*(e). \quad (39)$$

For each element of the small set $\pi_0(VX)$, choose a map f representing a preimage under α_X , and take their coproduct. Equation (39) proves the required surjectivity for (38). Its source and target are connective, and its map on π_0 is surjective, so the long exact sequence of its fiber shows that this fiber is connective. $\square$

4.5 A convergent tower with finite stages

Proposition 4.7. *For every $X \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq 0}$ there is an actual sequential diagram $(Y_n)_{n \geq 0}$ in $\mathrm{Cell}_{\mathrm{cyc}}$, together with a natural transformation to the constant diagram at X , such that*

$$\mathrm{colim}_{n \geq 0} Y_n \xrightarrow{\simeq} X.$$

One can take $Y_{-1} = 0$ and choose small sets I_n with cofiber sequences

$$Y_{n-1} \longrightarrow Y_n \longrightarrow \Sigma^n \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I_n)} \quad (40)$$

and with $\mathrm{cofib}(Y_n \rightarrow X) \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq n+1}$.

Proof. Set $K_0 = X$. Recursively apply Lemma 4.6 to choose maps

$$a_n : \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I_n)} \longrightarrow K_n, \quad K_{n+1} = \mathrm{fib}(a_n) \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq 0}.$$

Suppose that $Y_{n-1} \rightarrow X$ has been constructed with a specified cofiber sequence

$$Y_{n-1} \longrightarrow X \longrightarrow \Sigma^n K_n.$$

For $n = 0$ this is the sequence $0 \rightarrow X \xrightarrow{\mathrm{id}} X$. Define Y_n by the pullback square in $\mathrm{CycSp}_{\mathrm{NS}}$

$$\begin{array}{ccc} Y_n & \longrightarrow & \Sigma^n \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I_n)} \\ \downarrow & & \downarrow \Sigma^n a_n \\ X & \longrightarrow & \Sigma^n K_n. \end{array} \quad (41)$$

Its horizontal fibers satisfy

$$\mathrm{fib}(Y_n \rightarrow \Sigma^n \widetilde{\mathrm{THH}}(\mathbb{S}[t])^{(I_n)}) \simeq \mathrm{fib}(X \rightarrow \Sigma^n K_n) \simeq Y_{n-1}.$$

This identifies a map $Y_{n-1} \rightarrow Y_n$ over X and proves (40). Since $\mathrm{CycSp}_{\mathrm{NS}}$ is stable, the square is also a pushout. The cofibers of its vertical maps agree, giving

$$\mathrm{cofib}(Y_n \rightarrow X) \simeq \mathrm{cofib}(\Sigma^n a_n) \simeq \Sigma^{n+1} K_{n+1}. \quad (42)$$

The associated cofiber map $X \rightarrow \Sigma^{n+1} K_{n+1}$ is the map used at the next stage.

These constructions can be made in the diagram categories using the pullback and cofiber functors, with their universal cones. At each stage they extend the previously constructed finite diagram over X . Induction in the slice ∞ -category $(\mathrm{CycSp}_{\mathrm{NS}})_X$ therefore gives a sequential diagram there, or equivalently a diagram (Y_n) with a coherent natural transformation to the constant diagram at X . All fiber and cofiber calculations above take place in the stable category $\mathrm{CycSp}_{\mathrm{NS}}$.

Equation (40) implies inductively that each Y_n belongs to $\mathrm{Cell}_{\mathrm{cyc}}$: only $n+1$ successive cell attachments have been made at this stage. Equation (42) gives the stated connectivity of the error. Finally, V is exact and preserves colimits, so

$$V \mathrm{cofib}(\mathrm{colim}_n Y_n \rightarrow X) \simeq \mathrm{colim}_n \Sigma^{n+1} V K_{n+1} \simeq 0.$$

For the last equivalence, fix an integer k . Once $n+1 > k$, the k th homotopy group of $\Sigma^{n+1} V K_{n+1}$ is zero; homotopy groups of spectra commute with filtered colimits. The colimit therefore has every homotopy group zero. Conservativity of V proves the assertion. $\square$

Proof of Theorem 4.1. First suppose $X \in (\mathrm{CycSp}_{\mathrm{NS}})_{\geq 0}$ and take the sequential diagram of Proposition 4.7. The equivalence (34) induces an equivalence of diagram ∞ -categories

$$\mathrm{Fun}(\mathrm{NN}, \mathrm{Cell}_{\mathrm{mot}}) \xrightarrow{\simeq} \mathrm{Fun}(\mathrm{NN}, \mathrm{Cell}_{\mathrm{cyc}}). \quad (43)$$

Choose a preimage $\tilde{Y}_\bullet$ of $Y_\bullet$ under this equivalence, together with the resulting natural equivalence $\mathrm{THH}_{\mathrm{mot}} \tilde{Y}_\bullet \simeq Y_\bullet$. Thus all structure maps and their higher compatibilities are lifted at once. Form the colimit in the ambient category of motives:

$$M_X = \mathrm{colim}_n \tilde{Y}_n \quad \text{in } \mathrm{Mot}_{\mathrm{loc}}.$$

By (25) and Proposition 4.7,

$$\mathrm{THH}_{\mathrm{mot}}(M_X) \simeq \mathrm{colim}_n \mathrm{THH}_{\mathrm{mot}}(\tilde{Y}_n) \simeq \mathrm{colim}_n Y_n \simeq X.$$

Only the finite stages of this colimit are required to lie in $\mathrm{Cell}_{\mathrm{mot}}$.

If $VX \in \mathrm{Sp}_{\geq b}$, apply the connective case to $\Sigma^{-b}X$ and take $M = \Sigma^b M_{\Sigma^{-b}X}$. Exactness gives $\mathrm{THH}_{\mathrm{mot}}(M) \simeq X$. By [15, Theorem 1.5(1)], $U_{\mathrm{loc}} : \mathrm{Cat}^{\mathrm{perf}} \rightarrow \mathrm{Mot}_{\mathrm{loc}}$ is a Dwyer–Kan localization and in particular is essentially surjective. Choose $\mathcal{C}$ with $U_{\mathrm{loc}}(\mathcal{C}) \simeq M$. Then

$$\mathrm{THH}(\mathcal{C}) \simeq \mathrm{THH}_{\mathrm{mot}} U_{\mathrm{loc}}(\mathcal{C}) \simeq \mathrm{THH}_{\mathrm{mot}}(M) \simeq X.$$

Finally let $E \in \mathrm{CycSp}^{\mathrm{gen}}$ have bounded-below underlying spectrum. Apply the NS assertion to UE . By Theorem 3.1,

$$U\mathrm{THH}^{\mathrm{gen}}(\mathcal{C}) \simeq \mathrm{THH}(\mathcal{C}) \simeq UE.$$

Both genuine objects in this comparison have bounded-below underlying spectra. The equivalence (4) therefore lifts it to $\mathrm{THH}^{\mathrm{gen}}(\mathcal{C}) \simeq E$. $\square$

Remark 4.8. The connective rings of Lemma 4.2 provide targets for the mapping comparison (28). The final category $\mathcal{C}$ is instead obtained from the realization theorem for localizing motives. The argument places no connectivity condition on its mapping spectra and does not provide a presentation $\mathcal{C} \simeq \mathrm{Perf}(R)$ with R connective.

5 A cyclotomic spectrum with no genuine refinement

Fix a prime p and equip the Anderson dual sphere $I_{\mathbb{Z}}$ with the trivial $\mathbb{T}$ -action. Its defining fiber sequence and duality pairing are recalled in [Section A](#). We shall use the calculation

$$S_{p^2}(I_{\mathbb{Z}}) \simeq 0, \quad T_p^2(I_{\mathbb{Z}}) \not\simeq 0, \quad (44)$$

proved in [Lemma A.1](#). The first assertion concerns proper Tate fixed points; the second concerns two successive ordinary Tate constructions with their inherited actions.

5.1 The first composition constraint

For $Y \in \text{Fun}(B\mathbb{T}, \text{Sp})$, the proper Tate construction satisfies

$$Y^{\tau C_{p^2}} \simeq (Y^{hC_p})^{t(C_{p^2}/C_p)}. \quad (45)$$

Here and below the quotient acts through the extension $C_p \subset C_{p^2}$, and the displayed equivalence retains the residual $\mathbb{T}/C_{p^2}$ -action. The formula is [\[14, Example 1.24\(2\)\]](#). To see why the family of subgroups gives this expression, every proper subgroup of C_{p^2} lies in C_p . The inflation of $E(C_{p^2}/C_p)$ is consequently a classifying space for that family. In the isotropy-separation formula for geometric fixed points of the Borel completion of Y , taking C_p -fixed points first gives Y^{hC_p} ; separation by the inflated free C_{p^2}/C_p -space then gives its ordinary Tate construction. These operations commute with the residual action, giving [\(45\)](#) as an equivariant formula.

The root isomorphism identifies the subgroup used in the second application of T_p by

$$\rho_p(C_p) = C_{p^2}/C_p \subset \mathbb{T}/C_p.$$

If $\bar{\rho}_p : \mathbb{T}/C_p \xrightarrow{\sim} \mathbb{T}/C_{p^2}$ is the isomorphism induced by ρ_p , then $\bar{\rho}_p \rho_p = \rho_{p^2}$. Thus

$$T_p^2(Y) \simeq \rho_{p^2}^*((Y^{tC_p})^{t(C_{p^2}/C_p)}). \quad (46)$$

In particular, the second Tate construction uses the residual action coming from C_{p^2} .

The composition transformation for proper Tate constructions [\[14, Observation 1.25\]](#) gives

$$\alpha : S_{p^2} \longrightarrow T_p^2. \quad (47)$$

Before root reindexing it is the natural map

$$Y^{\tau C_{p^2}} \longrightarrow (Y^{\tau C_p})^{\tau(C_{p^2}/C_p)}.$$

Proper and ordinary Tate constructions agree for both prime-order groups on the right. The proper-Tate description of genuine cyclotomic spectra identifies this transformation with the comparison in the coherence for the pair (p, p) [\[1, Theorem B and Remark 0.8, diagram \(5\)\]](#).

Lemma 5.1. *Let $(Y, (\varphi_q)_q)$ be a Nikolaus–Scholze cyclotomic spectrum in $\text{EssIm}(U)$. There is a map $\sigma_{p^2} : Y \rightarrow S_{p^2}(Y)$ in $\text{Fun}(B\mathbb{T}, \text{Sp})$ such that*

$$T_p(\varphi_p)\varphi_p = \alpha_Y \sigma_{p^2} \quad \text{in } \pi_0 \text{Map}_{\text{Fun}(B\mathbb{T}, \text{Sp})}(Y, T_p^2 Y). \quad (48)$$

Proof. The proper-Tate description supplies, for a genuine cyclotomic spectrum with underlying homotopy circle spectrum Y , maps

$$\sigma_r : Y \longrightarrow S_r(Y), \quad r \geq 1,$$

and the composition coherences of [\[1, Theorem B and Remark 0.8\]](#). The prime-order map σ_p agrees with the Frobenius of U . Indeed, if E is the genuine spectrum and $e_p : \rho_p^* \Phi^{C_p} E \xrightarrow{\sim} E$

its cyclotomic structure equivalence, both maps are obtained by composing e_p^{-1} with the comparison from geometric fixed points to Tate fixed points. That comparison is induced by Borel completion [13, Propositions II.3.2 and II.3.4]. The (p, p) -coherence therefore gives the homotopy commutative square

$$\begin{array}{ccc} Y & \xrightarrow{\varphi_p} & T_p Y \\ \sigma_{p^2} \downarrow & & \downarrow T_p(\varphi_p) \\ S_{p^2} Y & \xrightarrow{\alpha_Y} & T_p^2 Y. \end{array}$$

This proves (48) on the image of U .

For invariance under equivalence, let $f : Y \xrightarrow{\sim} Y'$ be a cyclotomic equivalence with inverse g , and suppose Y' has a factorization through σ'_{p^2} . Cyclotomic compatibility and functoriality give

$$\varphi'_p f = T_p(f) \varphi_p, \quad T_p(\varphi'_p) \varphi'_p f = T_p^2(f) T_p(\varphi_p) \varphi_p.$$

Set $\sigma_{p^2} = S_{p^2}(g) \sigma'_{p^2} f$. Naturality of α then yields, in the homotopy category,

$$\begin{aligned} \alpha_Y \sigma_{p^2} &= T_p^2(g) \alpha_{Y'} \sigma'_{p^2} f \\ &= T_p^2(g) T_p(\varphi'_p) \varphi'_p f \\ &= T_p^2(gf) T_p(\varphi_p) \varphi_p = T_p(\varphi_p) \varphi_p. \end{aligned}$$

This proves the assertion on the essential image. $\square$

5.2 The shift Frobenius

Define the homotopy circle spectrum

$$X_{\text{And}, p} = I_{\mathbb{Z}} \oplus T_p I_{\mathbb{Z}} \oplus T_p^2 I_{\mathbb{Z}}. \quad (49)$$

The exact functor T_p preserves finite biproducts. Relative to the displayed decomposition and $T_p X_{\text{And}, p} = T_p I_{\mathbb{Z}} \oplus T_p^2 I_{\mathbb{Z}} \oplus T_p^3 I_{\mathbb{Z}}$, define

$$\varphi_p = \begin{pmatrix} 0 & \text{id}_{T_p I_{\mathbb{Z}}} & 0 \\ 0 & 0 & \text{id}_{T_p^2 I_{\mathbb{Z}}} \\ 0 & 0 & 0 \end{pmatrix} : X_{\text{And}, p} \longrightarrow T_p X_{\text{And}, p}. \quad (50)$$

For every prime $q \neq p$, set $\varphi_q = 0 : X_{\text{And}, p} \rightarrow T_q X_{\text{And}, p}$. All matrix entries are maps of homotopy circle spectra. These maps therefore define an object of CycSp_{NS} , whose definition imposes no composition constraint on the prime Frobenius maps [13, Definition II.1.6].

Theorem 5.2. *For every prime p , the cyclotomic spectrum $X_{\text{And}, p}$ of (49)–(50) does not belong to $\text{EssIm}(U)$. Consequently it is not equivalent to $\text{THH}_{\text{NS}}(\mathcal{C})$ for any $\mathcal{C} \in \text{Cat}^{\text{perf}}$.*

Proof. Let $i_j : T_p^j I_{\mathbb{Z}} \rightarrow X_{\text{And}, p}$ and $r_j : X_{\text{And}, p} \rightarrow T_p^j I_{\mathbb{Z}}$ be the inclusions and projections, for $0 \leq j \leq 2$. Formula (50) says

$$\varphi_p = T_p(i_0) r_1 + T_p(i_1) r_2.$$

After applying T_p and composing again with φ_p , the relations $r_a i_b = 0$ for $a \neq b$ and $r_a i_a = \text{id}$ give

$$T_p(\varphi_p) \varphi_p = T_p^2(i_0) r_2.$$

Indeed, among the four terms in the product, the only nonzero one contains $T_p(r_1 i_1) = \text{id}_{T_p^2 I_{\mathbb{Z}}}$. Hence

$$T_p^2(r_0) T_p(\varphi_p) \varphi_p i_2 = \text{id}_{T_p^2 I_{\mathbb{Z}}} \neq 0, \quad (51)$$

where nonvanishing follows from [Lemma A.1](#).

On the other hand, naturality of (47) gives

$$T_p^2(r_0)\alpha_{X_{\text{And},p}} = \alpha_{I_{\mathbb{Z}}}S_{p^2}(r_0) = 0, \quad (52)$$

because this map factors through $S_{p^2}(I_{\mathbb{Z}}) \simeq 0$ by the same appendix lemma. If $X_{\text{And},p}$ belonged to $\text{EssIm}(U)$, [Lemma 5.1](#) would identify $T_p(\varphi_p)\varphi_p$ with $\alpha_{X_{\text{And},p}}\sigma_{p^2}$. Precomposing with i_2 and postcomposing with $T_p^2(r_0)$ would contradict (51) and (52). Finally, [Theorem 3.1](#) places every $\text{THH}_{\text{NS}}(\mathcal{C})$ in $\text{EssIm}(U)$, proving the last assertion. $\square$

Remark 5.3. The obstruction involves only one prime and its first composition constraint. The underlying spectrum is necessarily unbounded below. More explicitly, [Corollary A.6](#) shows that $((I_{\mathbb{Z}})_{hC_p})^{t(C_{p^2}/C_p)} \not\cong 0$; the Tate orbit lemma [13, Lemma I.2.1] would force this spectrum to vanish if $I_{\mathbb{Z}}$ were bounded below. Since $I_{\mathbb{Z}}$ is a retract of $X_{\text{And},p}$, the latter is also unbounded below.

6 An obstruction across the primes

Theorem 6.1. *There are objects $X \in \text{CycSp}_{\text{NS}}$ and $E \in \text{CycSp}^{\text{gen}}$ with $UE \simeq X$ whose underlying homotopy circle spectrum is*

$$X \simeq H\mathbb{Z} \oplus \bigoplus_{p \text{ prime}} \Sigma^{-2p^2} H\mathbb{Z}, \quad (53)$$

with trivial circle action on every summand, and for which

$$X \notin \text{EssIm}(\text{THH}_{\text{NS}} : \text{Cat}^{\text{perf}} \longrightarrow \text{CycSp}_{\text{NS}}).$$

Consequently $E \notin \text{EssIm}(\text{THH}^{\text{gen}})$.

The obstruction compares the Frobenius maps for all primes on a single homotopy class. An integer-indexed May filtration gives a linear lower bound on the part of THH needed to produce these images. We shall choose the Frobenius of X so that its image in the p th summand is nonzero at depth $-2p^2$. The genuine lift will follow from vanishing results for proper and iterated Tate constructions on integral modules.

6.1 The integer May filtration

We regard $\mathbb{Z}^{\text{op}}$ as a symmetric monoidal category under addition; thus there is a morphism $i \rightarrow j$ precisely when $i \geq j$. Equip $\text{Fun}(\mathbb{Z}^{\text{op}}, \text{Sp})$ with Day convolution. For a filtered algebra, its Hochschild homology means the cyclic bar construction formed using this tensor product and then geometrically realized.

Lemma 6.2. *For every $\mathbb{E}_1$ -ring spectrum R , there is a decreasing filtration $F_{\bullet}\text{THH}(R)$ in $\text{Sp}^{B\mathbb{T}}$ such that*

$$F_n \text{THH}(R) \in \text{Sp}_{\geq n}^{B\mathbb{T}}, \quad (54)$$

$$\text{colim}_{n \rightarrow -\infty} F_n \text{THH}(R) \simeq \text{THH}(R). \quad (55)$$

For every prime p , the filtration admits maps

$$\varphi_{p,n} : F_n \text{THH}(R) \longrightarrow T_p(F_{pn} \text{THH}(R)) \quad (56)$$

natural in n and R . They fit into commutative squares in $\mathrm{Sp}^{B\mathbb{T}}$

$$\begin{array}{ccc} F_n \mathrm{THH}(R) & \xrightarrow{\varphi_{p,n}} & T_p(F_{pn} \mathrm{THH}(R)) \\ \downarrow & & \downarrow \\ \mathrm{THH}(R) & \xrightarrow{\varphi_p} & T_p \mathrm{THH}(R), \end{array} \quad (57)$$

where the bottom map is the standard cyclotomic Frobenius.

Proof. Put $W_n R = \tau_{\geq n} R$. We first construct the filtered algebra structure, including its coherent associativity. If $K \in \mathrm{Sp}_{\geq n}$, the truncation adjunction gives an equivalence of spaces

$$\mathrm{Map}(K, \tau_{\geq n} R) \xrightarrow{\sim} \mathrm{Map}(K, R). \quad (58)$$

For integers $i_1, \dots, i_r$ and $n \leq i_1 + \dots + i_r$, the spectrum $\bigotimes_a \tau_{\geq i_a} R$ is n -connective. Every r -ary multiplication of R therefore lifts through $\tau_{\geq n} R$, with a contractible space of choices. The same mapping-space equivalence lifts the homotopies comparing composites and all their higher coherences: after fixing an arity and the weights, these are simplices in the mapping spaces in (58). Since that equivalence is natural in the source, all operadic compositions and weight-transition maps are respected.

The Day unit has value $\mathbb{S}$ in weights $n \leq 0$ and zero in positive weights. The unit $\mathbb{S} \rightarrow R$ lifts through $\tau_{\geq 0} R$ and then through the structure maps to all weights $n \leq 0$. This treats the nullary operations as well. Hence $W_\bullet R$ is an $\mathbb{E}_1$ -algebra in the Day convolution category, naturally in R .

Define $F_\bullet \mathrm{THH}(R) = \mathrm{HH}(W_\bullet R)$. In simplicial degree r , its cyclic bar construction has weight n equal to

$$F_n B_r(R) = \mathrm{colim}_{i_0 + \dots + i_r \geq n} \tau_{\geq i_0} R \otimes \dots \otimes \tau_{\geq i_r} R. \quad (59)$$

The indexing category in (59) is the corresponding full subcategory of $(\mathbb{Z}^{\mathrm{op}})^{r+1}$. Each term is n -connective. Since $\mathrm{Sp}_{\geq n}$ is closed under colimits, both (59) and its geometric realization are n -connective. This proves (54).

Left Kan extension along the symmetric monoidal functor $\mathbb{Z}^{\mathrm{op}} \rightarrow *$ is the strong symmetric monoidal functor $\mathrm{colim}_{\mathbb{Z}^{\mathrm{op}}}$. For each k , the map $\pi_k \tau_{\geq n} R \rightarrow \pi_k R$ is an isomorphism once $n \leq k$; therefore $\mathrm{colim}_n W_n R \simeq R$. Applying this left Kan extension to the cyclic bar construction and using its preservation of geometric realizations proves (55), with its circle action.

For the Frobenius, we apply the twisted Tate diagonal of [2, Appendix A, Lemma A.7 and Corollary A.9]. Its hypothesis for a symmetric monoidal indexing category I is that

$$\mathrm{Map}_I(i^{\otimes p}, j)^{hC_p} \longrightarrow \mathrm{Map}_I(i^{\otimes p}, j)$$

be an equivalence. For $I = \mathbb{Z}^{\mathrm{op}}$, these spaces are either empty or contractible, so the hypothesis holds. Write L_p for left Kan extension along $i \mapsto pi$. The cited construction gives a circle-equivariant map $L_p F_\bullet \rightarrow T_p F_\bullet$, where Tate is computed pointwise. More explicitly,

$$(L_p F)_j \simeq \mathrm{colim}_{pi \geq j} F_i \simeq F_{[j/p]}. \quad (60)$$

The second equivalence uses the terminal object $[j/p]$ of the indexing category. Evaluating at $j = pn$ gives (56) with the residual circle action and the root reparametrization contained in T_p .

Finally, apply [2, Corollary A.15] to $\mathbb{Z}^{\mathrm{op}} \rightarrow *$. This compares the preceding construction with the ordinary Tate-diagonal cyclotomic structure on $\mathrm{THH}(R)$ and gives (57); see also [2, Example A.16]. The comparison uses the natural map from the colimit of the Tate constructions to the Tate construction of the colimit, obtained from the colimit cocone. $\square$

Theorem 6.3. *Let $\mathcal{C} \in \text{Cat}^{\text{perf}}$, set $Y = \text{THH}_{\text{NS}}(\mathcal{C})$, and let $a \in \pi_k VY$ for an integer k . There is an integer $N \geq 0$ such that, for every prime p ,*

$$\varphi_{p*}(a) \in \text{im}(\pi_k T_p(\tau_{\geq -Np} Y) \longrightarrow \pi_k T_p Y). \quad (61)$$

Here truncation is taken in the pointwise t -structure on $\text{Sp}^{B\mathbb{T}}$, and homotopy groups in the displayed formula refer to underlying spectra.

Proof. Choose a set of representatives for the objects of $\mathcal{C}$, and let S range over its finite subsets. Set

$$\mathcal{C}_S = \text{thick}_{\mathcal{C}}(S), \quad g_S = \bigoplus_{c \in S} c, \quad R_S = \text{End}_{\mathcal{C}}(g_S).$$

The mapping spectrum $\text{map}_{\mathcal{C}}(g_S, -)$ is a right R_S -module by precomposition. The resulting functor identifies $\mathcal{C}_S$ with $\text{Perf}(R_S)$: it is fully faithful on the generator g_S , and full faithfulness extends to finite cofibers, shifts, and retracts in both variables. Its image is exactly the thick subcategory generated by the free module R_S , namely $\text{Perf}(R_S)$.

The inclusions $\mathcal{C}_S \subset \mathcal{C}_{S'}$ form a filtered diagram of full stable subcategories. Every object of $\mathcal{C}$ occurs in one of them, and any finite collection of objects occurs in a common stage, with its full mapping spaces. Thus $\text{colim}_S \mathcal{C}_S \simeq \mathcal{C}$ in Cat^{perf} . The underlying categorical THH preserves filtered colimits and is Morita invariant by [4, Lemma 10.1 and Proposition 10.2]. The functor THH_{NS} supplies cyclotomic maps from these stages to Y . The source comparison (24) and its Morita descent in Section 3 identify $\text{THH}_{\text{NS}}(\text{Perf}(R_S))$ with the cyclic-bar $\text{THH}(R_S)$ equipped with its standard Tate-diagonal Frobenius. This is an identification in CycSp_{NS} . Since $\Sigma^k \mathbb{S}$ is compact, the class a is therefore the image of a class $a_R \in \pi_k \text{THH}(R)$ under a cyclotomic map

$$f : \text{THH}(R) \longrightarrow Y$$

for one of the rings R_S . Only this single stage is needed; the rings R_S need not form a diagram of unital ring maps.

By (55), a_R lifts to $\tilde{a} \in \pi_k F_{-N} \text{THH}(R)$ for some $N \geq 0$. This lift is chosen before selecting a prime. If $\iota_n : F_n \text{THH}(R) \rightarrow \text{THH}(R)$ is the filtration map, (57) gives

$$\varphi_{p*}(a) = (T_p(f \circ \iota_{-Np}))_*(\varphi_{p,-N})_*(\tilde{a}).$$

The source of $f \circ \iota_{-Np}$ is $(-Np)$ -connective. The truncation adjunction in $\text{Sp}^{B\mathbb{T}}$ consequently factors this map as

$$F_{-Np} \text{THH}(R) \longrightarrow \tau_{\geq -Np} Y \longrightarrow Y.$$

Applying T_p proves (61) for the same N at every prime. $\square$

6.2 Integral modules and proper Tate constructions

We identify $D(\mathbb{Z})$ with $\text{Mod}_{H\mathbb{Z}}(\text{Sp})$. Limits, colimits, and norm maps in this module category agree under the forgetful functor with the corresponding constructions on spectra [13, Section I.1]. In particular, finite-group Tate constructions may be performed inside $D(\mathbb{Z})$ while retaining all residual actions.

Lemma 6.4. *For every prime p and integer $d \geq 0$, there is an $H\mathbb{Z}$ -linear map in $\text{Sp}^{B\mathbb{T}}$*

$$b_{p,d} : H\mathbb{Z} \longrightarrow T_p(\Sigma^{-2d} H\mathbb{Z})$$

whose underlying map sends $1 \in \pi_0 H\mathbb{Z}$ to the nonzero generator

$$1 \in \pi_0 T_p(\Sigma^{-2d} H\mathbb{Z}) = \pi_{2d}(H\mathbb{Z}^{tC_p}) \cong \mathbb{F}_p.$$

The source and $\Sigma^{-2d} H\mathbb{Z}$ have trivial circle action before applying T_p .

Proof. The canonical map and computation in [13, Lemma II.4.2 and its proof] give

$$H\mathbb{Z}^{t\mathbb{T}} \longrightarrow (T_p H\mathbb{Z})^{h\mathbb{T}} \simeq \lim_n H\mathbb{Z}^{tC_{p^n}}.$$

For trivial integral coefficients, the homotopy groups of the terms on the right are $\mathbb{Z}/p^n$ in every even degree and zero in odd degrees. The transition maps are reduction maps. The corresponding calculation for the circle gives $\mathbb{Z}$ in even degrees and zero in odd degrees. The identification of the restriction maps in all degrees is supplied by [13, Lemma I.4.4]. The inverse systems of even homotopy groups are surjective, so their first derived limits vanish. Consequently, in degree $2d$, the composite to the underlying Tate spectrum is

$$\mathbb{Z} \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{F}_p, \quad 1 \longmapsto 1 \longmapsto 1. \quad (62)$$

In the stable category $D(\mathbb{Z})^{B\mathbb{T}}$, the mapping spectrum out of the trivially acting unit satisfies

$$\mathrm{map}_{D(\mathbb{Z})^{B\mathbb{T}}}(H\mathbb{Z}, \Sigma^{-2d} T_p H\mathbb{Z}) \simeq \Sigma^{-2d} (T_p H\mathbb{Z})^{h\mathbb{T}}.$$

The class $1 \in \mathbb{Z}_p$ in (62) therefore determines the asserted equivariant module map. Exactness of Tate identifies its target with $T_p(\Sigma^{-2d} H\mathbb{Z})$. $\square$

Lemma 6.5. *For $Y \in D(\mathbb{Z})^{B\mathbb{T}}$, one has*

$$S_m Y \simeq 0 \quad \text{if } m \text{ is composite,} \quad (63)$$

$$T_q T_p Y \simeq 0 \quad \text{for any primes } p, q. \quad (64)$$

These conclusions impose no boundedness condition on Y .

Proof. Fix a composite m . Let β_{C_m} be the Borel-completion right adjoint from spectra with C_m -action to genuine C_m -spectra. By [11, Proposition 5.24], the Borel-equivariant spectrum $\beta_{C_m} H\mathbb{Z}$ lies in the thick tensor ideal generated by $\Sigma_{C_m}^\infty (C_m/H)_+$ for elementary abelian subgroups $H \leq C_m$. A cyclic elementary abelian group is either trivial or of prime order. Since m is composite, all these subgroups are proper. Geometric fixed points Φ^{C_m} are exact and symmetric monoidal, and they send every such orbit spectrum to zero. Their kernel is a thick tensor ideal. It follows that

$$\Phi^{C_m} \beta_{C_m} H\mathbb{Z} \simeq 0. \quad (65)$$

The restriction of Y to C_m is a module over the trivially acting $H\mathbb{Z}$. The right adjoint β_{C_m} is lax symmetric monoidal, so $\beta_{C_m} Y$ is a module over $\beta_{C_m} H\mathbb{Z}$. Applying Φ^{C_m} shows that the proper Tate spectrum

$$Y^{\tau C_m} = \Phi^{C_m} \beta_{C_m} Y$$

is a unital module over the zero ring in (65). Its identity therefore factors through zero by the unit axiom, and the module is zero. The underlying spectrum detects the zero object in the residual action category. Root reparametrization now proves (63). This is the implication recorded in [1, Remark 2.18]; the module argument establishes its applicability to the possibly unbounded Y used here.

For (64) with $q = p$, restrict Y to C_{p^2} . The Tate fixpoint lemma applied to the bounded-above spectrum $H\mathbb{Z}$ gives

$$(H\mathbb{Z}^{hC_p})^{t(C_{p^2}/C_p)} \simeq 0.$$

Both

$$(Y_{hC_p})^{t(C_{p^2}/C_p)} \quad \text{and} \quad (Y^{hC_p})^{t(C_{p^2}/C_p)}$$

are unital modules over this ring. For homotopy fixed points this follows from their lax monoidal structure. For homotopy orbits, let $f : BC_{p^2} \rightarrow B(C_{p^2}/C_p)$ be the quotient map. The counit $f^* f_* H\mathbb{Z} \rightarrow H\mathbb{Z}$ makes Y a module over $f^* f_* H\mathbb{Z}$, and the projection formula

$$f_!(f^* f_* H\mathbb{Z} \otimes Y) \simeq f_* H\mathbb{Z} \otimes f_! Y$$

induces the $H\mathbb{Z}^{hC_p}$ -module structure on Y_{hC_p} with its quotient action. Applying the lax monoidal Tate functor yields the asserted module structure. These are precisely the module structures used in [13, Example I.2.3, footnote 9]. Thus both displayed spectra vanish for arbitrary Y . Apply the exact functor $(-)^{t(C_{p^2}/C_p)}$ to the residual-equivariant norm cofiber sequence

$$Y_{hC_p} \longrightarrow Y^{hC_p} \longrightarrow Y^{tC_p}.$$

The third term also vanishes. Under the root reparametrizations this third term is $T_p T_p Y$.

If $q \neq p$, the object $T_p Y$ is a module over $T_p H\mathbb{Z}$, whose underlying homotopy ring is $\mathbb{F}_p[u, u^{-1}]$, with $|u| = 2$. Multiplication by q is consequently an equivalence on $T_p Y$. The norm for C_q is an equivalence when q acts invertibly [13, Lemma I.2.8]; hence $T_q T_p Y \simeq 0$. $\square$

Proposition 6.6. *Let $Y \in D(\mathbb{Z})^{B\mathbb{T}}$, and fix maps $\psi_p : Y \rightarrow T_p Y$ in $\mathrm{Sp}^{B\mathbb{T}}$ for all primes p . The resulting NS cyclotomic spectrum admits a genuine lift. In the proper-Tate description of genuine cyclotomic spectra, the space of extensions of this fixed object and these fixed prime maps to a full structure is contractible.*

Proof. We use the equivalence of [1, Theorem 2.81] between genuine cyclotomic spectra and the right-lax limit of the proper-Tate left-lax action. Its functor at $m \in \mathbb{N}_{>0}^\times$ is S_m , with $S_1 = \mathrm{id}$ and $S_p = T_p$ for primes. The comparison transformations have the direction

$$\theta_{m,n} : S_{mn} \longrightarrow S_n S_m.$$

A structure consists of maps $\sigma_m : Y \rightarrow S_m Y$, with $\sigma_1 = \mathrm{id}$, and coherent comparisons between their composites and these action transformations.

Set $\sigma_p = \psi_p$ for primes. If m is composite, the space of choices of σ_m is $\mathrm{Map}_{\mathrm{Sp}^{B\mathbb{T}}}(Y, S_m Y) \simeq *$ by (63). For two nonunit integers m, n , the first compatibility is a filling of

$$\begin{array}{ccc} Y & \xrightarrow{\sigma_n} & S_n Y \\ \sigma_{mn} \downarrow & & \downarrow S_n(\sigma_m) \\ S_{mn} Y & \xrightarrow[\theta_{m,n}(Y)]{} & S_n S_m Y. \end{array} \quad (66)$$

The lower-right vertex is zero: a composite factor gives (63), and two prime factors give (64). Thus the two boundary composites in (66) lie in a contractible mapping space.

To construct all the higher data, use the normalized nerve of the one-object category $B\mathbb{N}_{>0}^\times$. A nondegenerate r -simplex is a word $w = (m_1, \dots, m_r)$ with every $m_i > 1$. Faces compose adjacent factors or remove an end factor; degeneracies insert 1. In the subdivision description of the right-lax limit, such a word carries the coherent r -cube of [1, Definition 1.18, Example 1.20, and Remarks 0.8–0.9], with final vertex

$$S_{m_r} \cdots S_{m_1} Y. \quad (67)$$

For $r \geq 2$, this vertex is zero. Indeed, if all factors are prime, the first two applications vanish by (64). If a composite factor occurs, its application vanishes by (63); before that application, the intermediate objects remain integral modules, since finite-group Tate constructions can be formed in $D(\mathbb{Z})$. Subsequent functors preserve zero.

Here is a relative section model for the space of choices. Put $B = (B\mathbb{N}_{>0}^\times)^{\mathrm{op}}$, and let $\mathcal{E} \rightarrow B$ be the locally cocartesian fibration encoding the proper-Tate action. For $r \geq 1$, let $D_r \subseteq \mathrm{sd}(B)$ be the full subcategory on words of nonunit length at most r , with the cocartesian arrows inherited from the subdivision augmentation $\mathrm{sd}(B) \rightarrow B$. Let $D_P \subseteq D_1$ consist of the empty word and the one-letter prime words. Write

$$\mathcal{S}_r = \mathrm{Fun}_{/B}^{\mathrm{cocart}}(D_r, \mathcal{E}), \quad \mathcal{S}_P = \mathrm{Fun}_{/B}^{\mathrm{cocart}}(D_P, \mathcal{E}).$$

The superscript requires preservation of the indicated inherited cocartesian arrows. By [1, Definition 1.18 and Example 1.20], a section over D_P is the data of an object and its prime maps. Let s_P be the section specified by $(Y, (\psi_p)_p)$, and define

$$\mathcal{K}_r = \text{hofib}_{s_P}(\mathcal{S}_r^\sim \longrightarrow \mathcal{S}_P^\sim).$$

Thus the underlying object and prime maps come with their specified identifications throughout. This length truncation retains every face of a retained word, since taking a face either deletes an end factor or multiplies adjacent factors and hence lowers the nonunit length. The union of the D_r is the full subdivision. The relative section space over s_P is therefore recovered by the inverse limit of the $\mathcal{K}_r$.

For $r = 1$, the remaining choices are maps to $S_m Y \simeq 0$ for composite m , so $\mathcal{K}_1$ is contractible. Fix a point of $\mathcal{K}_{r-1}$, with $r \geq 2$. Unwinding the subdivision sections, as in [1, Example 1.20 and Remark 0.8], identifies the new data with one coherent cube for each length- r word, relative to its prescribed faces. For a word w , this cube has parameter simplicial set $Q_r = N([1]^r) = (\Delta^1)^r$. Let $L_r \subset Q_r$ be the union of its coordinate boundary faces. The diagram on L_r is already specified: its faces use shorter words, their images under the S_m , and the fixed coherences of the action transformations θ . On intersections these prescriptions agree with the coherences already present in the shorter-word sections and in the action.

To verify the relative extension assertion, use a homotopy-coherent diagram model with Kan mapping spaces. A nondegenerate simplex of Q_r not contained in L_r is a strictly increasing chain of vertices in which every coordinate takes both values 0 and 1. Its first vertex is therefore $(0, \dots, 0)$ and its last is $(1, \dots, 1)$. Adjoin these simplices in increasing dimension. For a new d -simplex, all proper faces have already been specified, and its remaining coherence is a map from $(\Delta^1)^{d-1}$ into

$$M_w = \text{Map}_{\text{Sp}^{BT}}(Y, S_{m_r} \cdots S_{m_1} Y) \simeq *$$

with prescribed restriction to $\partial(\Delta^1)^{d-1}$. For $d = 1$ this boundary is empty. Since M_w is a contractible Kan complex, the restriction on these mapping spaces is a trivial fibration. Each relative filling space is consequently contractible; the same argument applies to families of fillings. It proves that the space of extensions from L_r to Q_r , relative to its entire specified boundary, is contractible.

Distinct length- r words attach along their shorter-word faces, on which the data have already been fixed. The preceding relative construction therefore fills their common-face compatibilities as well. Words containing 1 are recovered by the unit and degeneracy maps and add no choices. It follows that every restriction $\mathcal{K}_r \rightarrow \mathcal{K}_{r-1}$ is an equivalence of nonempty spaces. Hence $\lim_r \mathcal{K}_r$ is contractible and supplies the full right-lax structure.

The AMGR equivalence now gives a genuine cyclotomic object. Its prime structure maps under U are the prescribed ψ_p , so it lifts the specified NS object. $\square$

6.3 The counterexample and its bounded-below sources

Proof of Theorem 6.1. In $D(\mathbb{Z})^{BT}$, form

$$X = H\mathbb{Z} \oplus \bigoplus_{p \text{ prime}} \Sigma^{-2p^2} H\mathbb{Z},$$

with trivial circle action on its summands. Let $r_0 : X \rightarrow H\mathbb{Z}$, $i_p : \Sigma^{-2p^2} H\mathbb{Z} \rightarrow X$, and $r_p : X \rightarrow \Sigma^{-2p^2} H\mathbb{Z}$ be the projections and inclusions. By Lemma 6.4, there is an equivariant map $b_{p,p^2} : H\mathbb{Z} \rightarrow T_p(\Sigma^{-2p^2} H\mathbb{Z})$. Define

$$\varphi_p = T_p(i_p) \circ b_{p,p^2} \circ r_0 : X \longrightarrow T_p X. \quad (68)$$

These maps make X an object of CycSp_{NS} . If $x \in \pi_0 V X \cong \mathbb{Z}$ is the class of 1 in the first summand, then $r_p i_p = \text{id}$ gives

$$(T_p r_p)_* \varphi_{p*}(x) = 1 \in \pi_0 T_p(\Sigma^{-2p^2} H\mathbb{Z}) \cong \mathbb{F}_p. \quad (69)$$

Suppose that $X \simeq \text{THH}_{\text{NS}}(\mathcal{C})$. The condition in [Theorem 6.3](#) is invariant under cyclotomic equivalence, by naturality of truncation and T_p . Hence it holds for x and some $N \geq 0$. Choose a prime $p > N/2$, so that $-2p^2 < -Np$. The composite

$$\tau_{\geq -Np} X \longrightarrow X \xrightarrow{r_p} \Sigma^{-2p^2} H\mathbb{Z}$$

is null in $\text{Sp}^{B\mathbb{T}}$ by the orthogonality of its pointwise t-structure: its source is $(-Np)$ -connective and its target is concentrated in degree $-2p^2$. Applying T_p to this nullhomotopy and using (61) gives $(T_p r_p)_* \varphi_{p*}(x) = 0$, contrary to (69). Thus X is outside the image of THH_{NS} .

By [Proposition 6.6](#), the maps (68) extend to a genuine object E with $UE \simeq X$. If $E \simeq \text{THH}^{\text{gen}}(\mathcal{C})$, the natural comparison of [Theorem 3.1](#) would give $X \simeq \text{THH}_{\text{NS}}(\mathcal{C})$, which was just excluded. $\square$

Proposition 6.7. *For the object X of [Theorem 6.1](#), every cyclotomic map $f : Z \rightarrow X$ from a bounded-below NS cyclotomic spectrum induces the zero map $\pi_0 VZ \rightarrow \pi_0 VX$.*

Proof. Choose b with $VZ \in \text{Sp}_{\geq b}$. If $a \in \pi_0 VZ$ had nonzero image $m \in \mathbb{Z} \cong \pi_0 VX$, choose a prime p not dividing m and sufficiently large that $-2p^2 < b$. The map $r_p f : Z \rightarrow \Sigma^{-2p^2} H\mathbb{Z}$ is null in $\text{Sp}^{B\mathbb{T}}$ by t-structure orthogonality. Cyclotomic compatibility and (69) nevertheless give

$$0 = (T_p(r_p f))_* \varphi_{p*}^Z(a) = (T_p r_p)_* \varphi_{p*}^X(f_* a) = m \bmod p \neq 0,$$

a contradiction. $\square$

Corollary 6.8. *The object X is not a filtered colimit of bounded-below objects of CycSp_{NS} . Its genuine lift E is not a filtered colimit of bounded-below objects of $\text{CycSp}^{\text{gen}}$.*

Proof. By [Lemma 2.1](#), $V : \text{CycSp}_{\text{NS}} \rightarrow \text{Sp}$ preserves colimits. If $X \simeq \text{colim}_i X_i$ were a filtered presentation by bounded-below objects, compactness of $\mathbb{S}$ would make $x \in \pi_0 VX$ the image of a class in some $\pi_0 VX_i$. This contradicts [Proposition 6.7](#).

The genuine underlying-spectrum functor VU also preserves colimits by [Lemma 2.1](#). If $E \simeq \text{colim}_i E_i$ were a filtered colimit of bounded-below genuine objects, a class in some $\pi_0 VUE_i$ would map to $x \in \pi_0 VUE$. The induced NS cyclotomic map $UE_i \rightarrow UE \simeq X$ again contradicts [Proposition 6.7](#). $\square$

7 Essential images and the role of boundedness

Proof of [Theorem 1.1](#). The construction and natural comparison in part (i) are [Theorem 3.1](#). Part (ii) is [Theorem 4.1](#), together with the bounded-below comparison (4). The natural equivalence in part (i) gives

$$\text{EssIm}(\text{THH}_{\text{NS}}) \subseteq \text{EssIm}(U).$$

The object of [Theorem 5.2](#) lies outside $\text{EssIm}(U)$, proving the second strict inclusion in (1). The object of [Theorem 6.1](#) belongs to $\text{EssIm}(U)$ and does not belong to $\text{EssIm}(\text{THH}_{\text{NS}})$, proving the first. If its genuine refinement were equivalent to $\text{THH}^{\text{gen}}(\mathcal{C})$, applying U and part (i) would give precisely the NS realization excluded by that theorem. Hence THH^{gen} is not essentially surjective. $\square$

The May estimate has no obstruction on a bounded-below target. Indeed, if $VX \in \mathrm{Sp}_{\geq b}$, choose an integer $N \geq 0$ with $-2N \leq b$. For every prime p we then have $-Np \leq b$, and $\tau_{\geq -Np}X \rightarrow X$ is an equivalence of homotopy circle spectra. Thus the factorization required in [Theorem 6.3](#) is automatic. The positive realization theorem supplies the converse existence statement by lifting a cellular diagram in motives, rather than by constructing cyclotomic connective covers of an arbitrary unbounded object.

Two elementary calculations specify further limits of such reconstruction arguments. They also explain the use of equivariant connective covers and of finite stages in the preceding proofs.

Proposition 7.1. *For a prime p and an integer $d > 0$, with trivial actions on source and target, one has*

$$[\mathbb{S}, \Sigma^{2d}H\mathbb{Z}]_{\mathrm{Sp}_{BC_p}} \cong \mathbb{Z}/p, \quad [\mathbb{S}, \Sigma^{2d}H\mathbb{Z}]_{\mathrm{Sp}} = 0.$$

Proof. The equivariant mapping spectrum is $(\Sigma^{2d}H\mathbb{Z})^{hC_p}$, whose degree-zero homotopy group is $H^{2d}(BC_p; \mathbb{Z})$. Applying $\mathrm{Hom}_{\mathbb{Z}[C_p]}(-, \mathbb{Z})$ to the periodic resolution with differentials $1 - g$ and $1 + g + \cdots + g^{p-1}$ gives alternating differentials 0 and p . Its positive even cohomology is $\mathbb{Z}/p$. The nonequivariant group is $\pi_{-2d}H\mathbb{Z} = 0$. $\square$

In particular, a nonequivariant nullhomotopy cannot be substituted for the equivariant nullhomotopy in [\(3\)](#).

Proposition 7.2. *The functor $\mathrm{TC} : \mathrm{CycSp}_{\mathrm{NS}} \rightarrow \mathrm{Sp}$ does not preserve filtered colimits, even on rational cyclotomic spectra with trivial circle action.*

Proof. Every finite-group Tate construction of a rational spectrum vanishes [[13](#), Lemma I.2.8]. Such a circle spectrum therefore has a unique NS cyclotomic structure. The defining fiber sequence for TC [[13](#), Proposition II.1.9] reduces to $\mathrm{TC}(Z) \simeq Z^{h\mathbb{T}}$ on these objects.

For $n \geq 0$, set

$$Z_n = \bigoplus_{i=0}^n \Sigma^{2i}H\mathbb{Q}, \quad Z_\infty = \mathrm{colim}_n Z_n = \bigoplus_{i \geq 0} \Sigma^{2i}H\mathbb{Q},$$

with the standard inclusions and trivial actions. Since $B\mathbb{T} \simeq \mathbb{CP}^\infty$, its rational suspension spectrum, viewed as an $H\mathbb{Q}$ -module, is

$$H\mathbb{Q} \otimes \Sigma_+^\infty B\mathbb{T} \simeq \bigoplus_{j \geq 0} \Sigma^{2j}H\mathbb{Q}.$$

For example this follows from the cellular chain complex, with one rational cell in each nonnegative even degree and zero differential. The trivial-action homotopy fixed points of a rational spectrum Z are therefore

$$Z^{h\mathbb{T}} \simeq \mathrm{map}_{\mathrm{Mod}_{H\mathbb{Q}}}(H\mathbb{Q} \otimes \Sigma_+^\infty B\mathbb{T}, Z), \quad \pi_0 Z^{h\mathbb{T}} \cong \prod_{j \geq 0} \pi_{2j} Z.$$

Consequently the natural comparison on degree-zero homotopy groups is

$$\mathrm{colim}_n \pi_0 \mathrm{TC}(Z_n) = \bigoplus_{i \geq 0} \mathbb{Q} \longrightarrow \prod_{i \geq 0} \mathbb{Q} = \pi_0 \mathrm{TC}(Z_\infty).$$

It is the inclusion of finitely supported sequences and is not surjective. $\square$

The proof of bounded-below realization uses the colimit-preserving motivic extension of THH. The required mapping comparisons are obtained on the finite stable closure of arbitrary coproducts by a retract argument. [Proposition 7.2](#) explains why filtered-colimit preservation of a TC-valued auxiliary construction cannot replace that argument without a separate proof.

A Anderson duality and iterated Tate constructions

For an injective abelian group M , let $I_M\mathbb{S}$ denote the Brown–Comenetz spectrum characterized by $[Z, I_M\mathbb{S}] = \text{Hom}_{\mathbb{Z}}(\pi_0 Z, M)$. The Anderson dual of the sphere is

$$I_{\mathbb{Z}} = \text{fib}(I_{\mathbb{Q}}\mathbb{S} \longrightarrow I_{\mathbb{Q}/\mathbb{Z}}\mathbb{S}), \quad D_{\mathbb{Z}}(Z) = F(Z, I_{\mathbb{Z}}).$$

The evaluation pairing $Z \otimes D_{\mathbb{Z}}(Z) \rightarrow I_{\mathbb{Z}}$ induces the natural double-dual map $Z \rightarrow D_{\mathbb{Z}}^2(Z)$. If Z carries a group action, $D_{\mathbb{Z}}(Z)$ has the contragredient action: g acts by precomposition with g^{-1} . Thus evaluation is equivariant when $I_{\mathbb{Z}}$ has trivial action. For the cyclic-group calculations below, $I_{\mathbb{Z}}$ has trivial $\mathbb{T}$ -action, and every residual C_{p^2}/C_p -action is induced from its restriction to C_{p^2} .

Lemma A.1. *For every prime p , the Anderson dual sphere satisfies*

$$S_{p^2}(I_{\mathbb{Z}}) \simeq 0, \quad T_p^2(I_{\mathbb{Z}}) \not\simeq 0. \quad (70)$$

The vanishing follows from the upper bound on $I_{\mathbb{Z}}$ and the Tate fixpoint lemma. For nonvanishing, we identify $D_{\mathbb{Z}}(I_{\mathbb{Z}}^{tC_p})$ with a shifted p -complete sphere, including its residual action. We then prove that $I_{\mathbb{Z}}^{tC_p}$ is bounded above, which permits a second application of Tate duality. Its nonzero result proves the second assertion of [Lemma A.1](#). This supplies the details of the Anderson-dual example in [\[13, Example I.2.3\(ii\)\]](#).

A.1 Duality on bounded-above spectra

The universal-coefficient sequence is

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}^1(\pi_{-n-1}Z, \mathbb{Z}) \longrightarrow \pi_n D_{\mathbb{Z}}(Z) \longrightarrow \text{Hom}_{\mathbb{Z}}(\pi_{-n}Z, \mathbb{Z}) \longrightarrow 0. \quad (71)$$

It follows by applying $F(Z, -)$ to the defining fiber sequence; see [\[9, Definitions 2.1–2.2 and Lemma 2.4\]](#). In particular,

$$Z \in \text{Sp}_{\leq d} \implies D_{\mathbb{Z}}(Z) \in \text{Sp}_{\geq -d-1}. \quad (72)$$

Lemma A.2. *The spectrum $I_{\mathbb{Z}}$ belongs to $\text{Sp}_{\leq 0}$, and the natural map $\mathbb{S} \rightarrow D_{\mathbb{Z}}(I_{\mathbb{Z}})$ is an equivalence, compatible with trivial C_{p^2} -actions.*

Proof. For $Z = \mathbb{S}$ and $n > 0$, both outside terms of (71) vanish, so $I_{\mathbb{Z}} = D_{\mathbb{Z}}(\mathbb{S})$ is bounded above by zero. Each homotopy group of $\mathbb{S}$ is finitely generated. The double-duality theorem [\[9, Theorem 2.9\]](#) therefore makes the evaluation map $\mathbb{S} \rightarrow D_{\mathbb{Z}}^2(\mathbb{S}) = D_{\mathbb{Z}}(I_{\mathbb{Z}})$ an equivalence. Naturality makes it equivariant for the specified trivial actions. $\square$

Lemma A.3. *Let G be a finite group and $B \in \text{Fun}(BG, \text{Sp})$. There is a natural equivalence*

$$D_{\mathbb{Z}}(B_{hG}) \simeq D_{\mathbb{Z}}(B)^{hG}. \quad (73)$$

If the underlying spectrum of B is bounded above, there are also natural equivalences

$$D_{\mathbb{Z}}(B^{hG}) \simeq D_{\mathbb{Z}}(B)_{hG}, \quad D_{\mathbb{Z}}(B^{tG}) \simeq \Sigma^{-1} D_{\mathbb{Z}}(B)^{tG}. \quad (74)$$

If G is normal in a finite group L and B has an L -action, all three equivalences respect the residual L/G -actions.

Proof. The equivalence (73) follows because $F(-, I_{\mathbb{Z}})$ takes colimits to limits. We construct the first map in (74) from evaluation, and prove it is an equivalence by a boundedness estimate.

For a finite free G -CW complex W , evaluation gives a pairing

$$(W_+ \otimes_G D_{\mathbb{Z}}(B)) \otimes F_G(W_+, B) \longrightarrow I_{\mathbb{Z}}.$$

Here F_G denotes the spectrum of equivariant maps, and the tensor over G is the derived balanced tensor. The pairing evaluates an equivariant function at the point of W and then uses $D_{\mathbb{Z}}(B) \otimes B \rightarrow I_{\mathbb{Z}}$. It descends over G because the actions are contragredient. The adjoint map is a natural equivalence

$$W_+ \otimes_G D_{\mathbb{Z}}(B) \xrightarrow{\sim} D_{\mathbb{Z}}(F_G(W_+, B)). \quad (75)$$

For a free cell with quotient sphere S^d , this is the equivalence $\Sigma^d D_{\mathbb{Z}}(B) \simeq D_{\mathbb{Z}}(\Omega^d B)$. Finite induction over the cell attachments proves the assertion: mapping out of an attachment gives a fiber sequence, applying $D_{\mathbb{Z}}$ reverses it to a cofiber sequence, and balanced tensor preserves the corresponding cofiber sequence. The induction uses the evaluation map itself, and hence is natural in W and B .

Choose the bar model for EG , with finite free n -skeleta E_n . Suppose $B \in \mathrm{Sp}_{\leq b}$. The restriction map has fiber

$$\mathrm{Fib}_n = \mathrm{fib}(B^{hG} \longrightarrow F_G(E_{n+}, B)) \simeq F_G(EG_+/E_{n+}, B).$$

All relative cells have dimension at least $n+1$, so mapping them into B gives spectra in $\mathrm{Sp}_{\leq b-n-1}$. This subcategory is closed under limits. Consequently

$$\mathrm{Fib}_n \in \mathrm{Sp}_{\leq b-n-1}, \quad D_{\mathbb{Z}}(\mathrm{Fib}_n) \in \mathrm{Sp}_{\geq n-b},$$

where the second assertion uses (72). Dualizing the fiber sequence gives

$$D_{\mathbb{Z}}(F_G(E_{n+}, B)) \longrightarrow D_{\mathbb{Z}}(B^{hG}) \longrightarrow D_{\mathbb{Z}}(\mathrm{Fib}_n).$$

The first map is thus an isomorphism on each fixed homotopy group for all sufficiently large n . The maps (75) commute with skeleton inclusions, so passing to the colimit proves

$$\begin{aligned} D_{\mathbb{Z}}(B)_{hG} &\simeq \mathrm{colim}_n (E_{n+} \otimes_G D_{\mathbb{Z}}(B)) \\ &\simeq \mathrm{colim}_n D_{\mathbb{Z}}(F_G(E_{n+}, B)) \xrightarrow{\sim} D_{\mathbb{Z}}(B^{hG}). \end{aligned} \quad (76)$$

The norm is self-transpose under these evaluation pairings. More precisely, the following square commutes:

$$\begin{array}{ccc} D_{\mathbb{Z}}(B)_{hG} & \xrightarrow{\mathrm{Nm}_{D_{\mathbb{Z}}(B)}} & D_{\mathbb{Z}}(B)^{hG} \\ \simeq \downarrow & & \simeq \downarrow \\ D_{\mathbb{Z}}(B^{hG}) & \xrightarrow{D_{\mathbb{Z}}(\mathrm{Nm}_B)} & D_{\mathbb{Z}}(B_{hG}). \end{array} \quad (77)$$

To verify this for the canonical norm, put $C = D_{\mathbb{Z}}(B)$, write $f : BG \rightarrow *$, and let $p_0, p_1 : BG \times BG \rightarrow BG$ be the projections. Let δ be the diagonal and s interchange the two factors. In [13, Construction I.1.7], the norm is the successive adjoint of the composite

$$\kappa_B : p_0^* B \xrightarrow{\eta} \delta_* B \xrightarrow{\mathrm{Nm}_{\delta}^{-1}} \delta_! B \xrightarrow{\varepsilon} p_1^* B.$$

Here η and ε are the unit and counit for $\delta^* \dashv \delta_*$ and $\delta_! \dashv \delta^*$, using $\delta^* p_i^* = \mathrm{id}$. More precisely, under the adjunctions and right base change $p_{0*} p_1^* \simeq f^* f_*$, the map κ_B corresponds to $B \rightarrow f^* f_* B$ and then to $\mathrm{Nm}_B : f_! B \rightarrow f_* B$.

The fibers of δ are finite sets. Pointwise Anderson duality therefore identifies $D_{\mathbb{Z}}(\delta_! B)$ with $\delta_* C$ and $D_{\mathbb{Z}}(\delta_* B)$ with $\delta_! C$. It exchanges the displayed unit and counit. The map Nm_{δ} is the finite biproduct identification, with identity diagonal entries and zero off-diagonal entries, so its inverse transposes to the same inverse for C . Thus

$$D_{\mathbb{Z}}(\kappa_B) = (p_1^* C \xrightarrow{\eta} \delta_* C \xrightarrow{\text{Nm}_{\delta}^{-1}} \delta_! C \xrightarrow{\varepsilon} p_0^* C) = s^* \kappa_C.$$

To identify its adjoints with the vertical maps in (77), use the evaluation pairings

$$(f_! C) \otimes f_* B \longrightarrow I_{\mathbb{Z}}, \quad f_* C \otimes (f_! B) \longrightarrow I_{\mathbb{Z}}.$$

They are induced, respectively, by the counits $f^* f_* B \rightarrow B$ and $f^* f_* C \rightarrow C$, followed by $C \otimes B \rightarrow I_{\mathbb{Z}}$. The first is exactly the pairing defining (76): its restriction to each finite skeleton is the point-evaluation pairing above. The second gives (73). Set $q = fp_0 = fp_1$. By

$$(f_! C) \otimes (f_! B) \simeq q_!(p_0^* C \otimes p_1^* B),$$

the two routes around (77) are adjoint to the following pairings on $BG \times BG$:

$$\begin{aligned} p_0^* C \otimes p_1^* B &\xrightarrow{\kappa_C \otimes \text{id}} p_1^* C \otimes p_1^* B \xrightarrow{p_1^* \text{ev}} q^* I_{\mathbb{Z}}, \\ p_0^* C \otimes p_1^* B &\xrightarrow{\text{id} \otimes s^* \kappa_B} p_0^* C \otimes p_0^* B \xrightarrow{p_0^* \text{ev}} q^* I_{\mathbb{Z}}. \end{aligned}$$

These agree by the transposition identity just proved. This establishes (77) with the specified evaluation identifications.

As a finite-fiber check of this construction, the transfer is the sum of the action maps indexed by G . Transposing $g : B \rightarrow B$ gives precomposition by g , which is the action of g^{-1} on $D_{\mathbb{Z}}(B)$. Hence

$$\sum_{g \in G} D_{\mathbb{Z}}(g) = \sum_{g \in G} g^{-1} = \sum_{g \in G} g \quad \text{on } D_{\mathbb{Z}}(B).$$

Equivalently, as an identity of maps $D_{\mathbb{Z}}(B) \otimes B \rightarrow I_{\mathbb{Z}}$,

$$\text{ev} \circ ((\sum_g g) \otimes \text{id}) = \text{ev} \circ (\text{id} \otimes (\sum_g g)).$$

Finally, applying $D_{\mathbb{Z}}$ to the norm cofiber sequence yields

$$\begin{aligned} D_{\mathbb{Z}}(B^{tG}) &\simeq \text{fib}(D_{\mathbb{Z}}(B^{hG}) \longrightarrow D_{\mathbb{Z}}(B_{hG})) \\ &\simeq \text{fib}(D_{\mathbb{Z}}(B)_{hG} \xrightarrow{\text{Nm}_{D_{\mathbb{Z}}(B)}} D_{\mathbb{Z}}(B)^{hG}) \\ &\simeq \Sigma^{-1} D_{\mathbb{Z}}(B)^{tG}. \end{aligned}$$

For residual equivariance, take finite free L -skeleta of EL and restrict them to G . Since L is finite these are still finite free G -CW complexes, and their union is a model for EG . The evaluation maps, restrictions, and transfers above carry the residual L/G -actions. In the transfer identity, the L -action permutes the summands by conjugation, which preserves their sum because G is normal. All the comparisons are therefore L/G -equivariant. The equivalences already proved on underlying spectra are equivalences in $\text{Fun}(B(L/G), \text{Sp})$. $\square$

A.2 The first Tate construction and its upper bound

Lemma A.4. *For the residual action induced by the trivial C_{p^2} -action on $I_{\mathbb{Z}}$, there is an equivalence*

$$D_{\mathbb{Z}}(I_{\mathbb{Z}}^{tC_p}) \simeq \Sigma^{-1} \mathbb{S}_p^{\wedge} \quad \text{in } \text{Fun}(B(C_{p^2}/C_p), \text{Sp}), \quad (78)$$

where the right-hand side has trivial action. Moreover, $I_{\mathbb{Z}}^{tC_p} \in \text{Sp}_{\leq 0}$.

Proof. Since $I_{\mathbb{Z}} \in \mathrm{Sp}_{\leq 0}$, [Lemmas A.2](#) and [A.3](#) give residual-equivariant equivalences

$$D_{\mathbb{Z}}(I_{\mathbb{Z}}^{tC_p}) \simeq \Sigma^{-1} D_{\mathbb{Z}}(I_{\mathbb{Z}})^{tC_p} \simeq \Sigma^{-1} \mathbb{S}^{tC_p}.$$

The Segal-conjecture calculation [[13](#), Example I.2.3(i)] identifies $\mathbb{S}^{tC_p}$ with $\mathbb{S}_p^{\wedge}$. The map from $\mathbb{S}$ giving this identification is C_{p^2}/C_p -equivariant with trivial action on its source, as specified in that example. Factoring it through the equivariant p -completion of the source identifies the residual action with the trivial action on $\mathbb{S}_p^{\wedge}$. This proves [\(78\)](#).

To apply [Lemma A.3](#) once more, we must prove an upper bound for $I_{\mathbb{Z}}^{tC_p}$. The Postnikov convergence statement [[13](#), Lemma I.2.6(ii)] gives

$$I_{\mathbb{Z}}^{tC_p} \simeq \operatorname{colim}_{r \geq 0} (\tau_{\geq -r} I_{\mathbb{Z}})^{tC_p}. \quad (79)$$

Each $\tau_{\geq -r} I_{\mathbb{Z}}$ has only finitely many nonzero Postnikov layers, because $I_{\mathbb{Z}} \leq 0$. For a C_p -module M , the Tate construction of HM has

$$\pi_k((HM)^{tC_p}) = \widehat{H}^{-k}(C_p; M).$$

These groups are killed by p . This follows from the complete cyclic resolution, whose differentials alternate between $g - 1$ and $1 + g + \cdots + g^{p-1}$ for a generator g of C_p : multiplication by p is zero on its cohomology, since $p - (1 + g + \cdots + g^{p-1})$ is divisible by $1 - g$ in $\mathbb{Z}[C_p]$.

Exactness of the Tate functor now shows by induction that the Tate construction of a spectrum with ℓ such Postnikov layers has homotopy groups killed by p^{ℓ} . Indeed, the long exact sequence for one additional layer presents each homotopy group as an extension of a subquotient from the preceding layers by a subquotient from the new layer. The exponents multiply across such an extension. By [\(79\)](#), each element of $\pi_m(I_{\mathbb{Z}}^{tC_p})$ comes from a finite stage, and hence

$$\pi_m(I_{\mathbb{Z}}^{tC_p}) \text{ is a } p\text{-primary torsion group for every } m. \quad (80)$$

For a torsion abelian group M , one has $\operatorname{Hom}(M, \mathbb{Z}) = 0$ and

$$\operatorname{Ext}^1(M, \mathbb{Z}) \cong \operatorname{Hom}(M, \mathbb{Q}/\mathbb{Z}).$$

The latter follows from the injective resolution $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ and $\operatorname{Hom}(M, \mathbb{Q}) = 0$. Thus [\(71\)](#) and [\(80\)](#) give

$$\pi_k D_{\mathbb{Z}}(I_{\mathbb{Z}}^{tC_p}) \cong \operatorname{Hom}(\pi_{-k-1}(I_{\mathbb{Z}}^{tC_p}), \mathbb{Q}/\mathbb{Z}). \quad (81)$$

The left side vanishes for $k < -1$ by [\(78\)](#). For $m > 0$, take $k = -m - 1$. Since $\mathbb{Q}/\mathbb{Z}$ is an injective cogenerator, [\(81\)](#) forces $\pi_m(I_{\mathbb{Z}}^{tC_p}) = 0$. Concretely, a nonzero element of a torsion group generates a finite cyclic subgroup admitting a nonzero map to $\mathbb{Q}/\mathbb{Z}$, and injectivity extends that map to the whole group. This proves the required upper bound. $\square$

A.3 The residual Tate construction

Lemma A.5. *With the residual C_{p^2}/C_p -action induced from $I_{\mathbb{Z}}$, there is an equivalence of spectra*

$$D_{\mathbb{Z}}((I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)}) \simeq \Sigma^{-2} \mathbb{S}_p^{\wedge}. \quad (82)$$

In particular, $(I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)}$ is nonzero.

Proof. The upper bound in Lemma A.4 permits the application of Lemma A.3 to $I_{\mathbb{Z}}^{tC_p}$ with its residual action. Together with (78), it yields

$$\begin{aligned} D_{\mathbb{Z}}((I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)}) &\simeq \Sigma^{-1} D_{\mathbb{Z}}(I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)} \\ &\simeq \Sigma^{-2} (\mathbb{S}_p^{\wedge})^{t(C_{p^2}/C_p)} \\ &\simeq \Sigma^{-2} \mathbb{S}_p^{\wedge}. \end{aligned}$$

For the last equivalence the action is trivial by Lemma A.4. The group C_{p^2}/C_p has order p . Applying [13, Lemma I.2.9] to the connective sphere shows that its Tate construction is unchanged by p -completion, and [13, Example I.2.3(i)] then identifies it with $\mathbb{S}_p^{\wedge}$. The right side of (82) has $\pi_{-2} = \mathbb{Z}_p \neq 0$. A zero spectrum has zero Anderson dual, so the spectrum on the left before dualization cannot vanish. $\square$

Corollary A.6. *The norm sequence for $I_{\mathbb{Z}}$, with its inherited quotient action, induces an equivalence*

$$(I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)} \simeq \Sigma((I_{\mathbb{Z}})_{hC_p})^{t(C_{p^2}/C_p)} \not\simeq 0. \quad (83)$$

Proof. The upper bound $I_{\mathbb{Z}} \leq 0$ in Lemma A.2 satisfies the hypothesis of the Tate fixpoint lemma [13, Lemma I.2.2], which gives

$$(I_{\mathbb{Z}}^{hC_p})^{t(C_{p^2}/C_p)} \simeq 0. \quad (84)$$

Apply the exact functor $(-)^{t(C_{p^2}/C_p)}$ to the C_{p^2}/C_p -equivariant norm cofiber sequence

$$(I_{\mathbb{Z}})_{hC_p} \xrightarrow{\text{Nm}_{I_{\mathbb{Z}}}} I_{\mathbb{Z}}^{hC_p} \longrightarrow I_{\mathbb{Z}}^{tC_p}.$$

Its middle term becomes zero by (84), giving the equivalence in (83). Nonvanishing is Lemma A.5. $\square$

Proof of Lemma A.1. The proper-Tate formula (45), together with (84), gives $S_{p^2}(I_{\mathbb{Z}}) \simeq 0$. Equation (46) identifies the underlying spectrum of $T_p^2(I_{\mathbb{Z}})$ with $(I_{\mathbb{Z}}^{tC_p})^{t(C_{p^2}/C_p)}$, which is nonzero by Lemma A.5. Root reindexing changes the action and leaves the underlying spectrum unchanged, proving both assertions. $\square$

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